

A mix within

Linear Mixed Models and their applications

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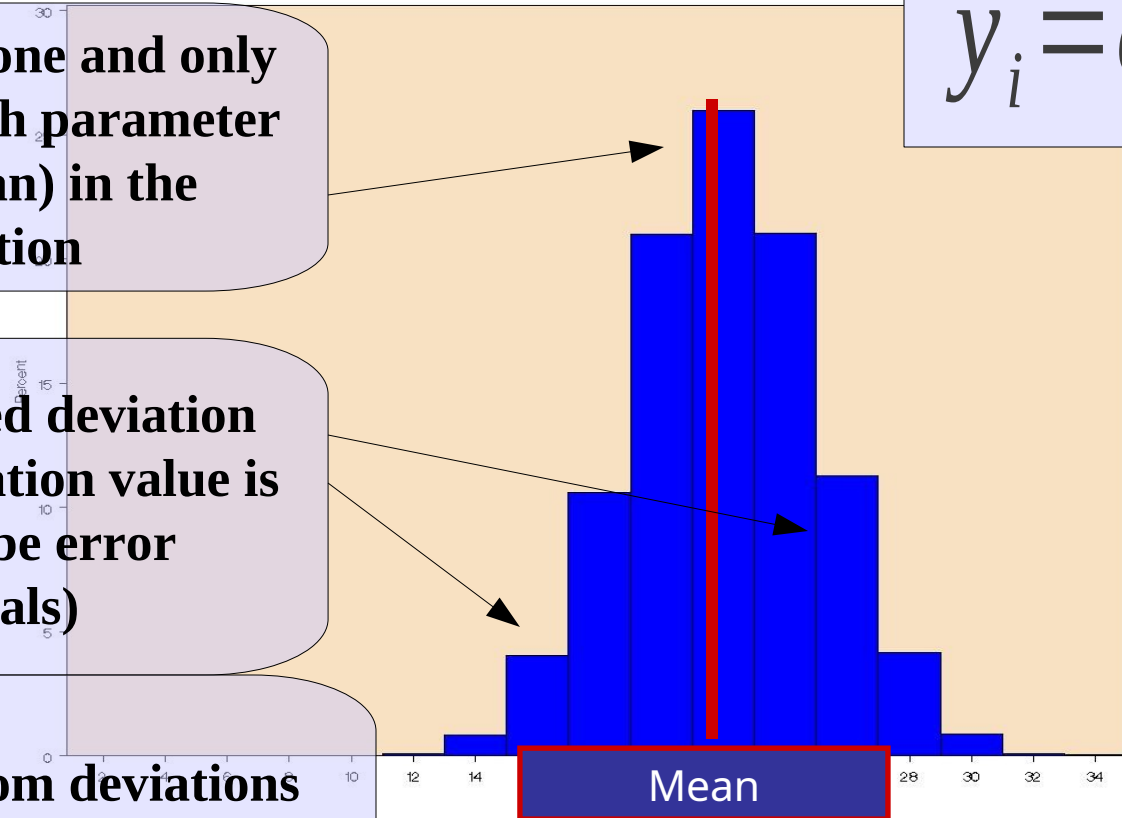
Some GLM Assumptions

1) There exists one and only one value of each parameter (e.g. the mean) in the population

2) Any observed deviation from the population value is deemed to be error (residuals)

3) The random deviations from the model are normally distributed

$$y_i = a + e_i$$

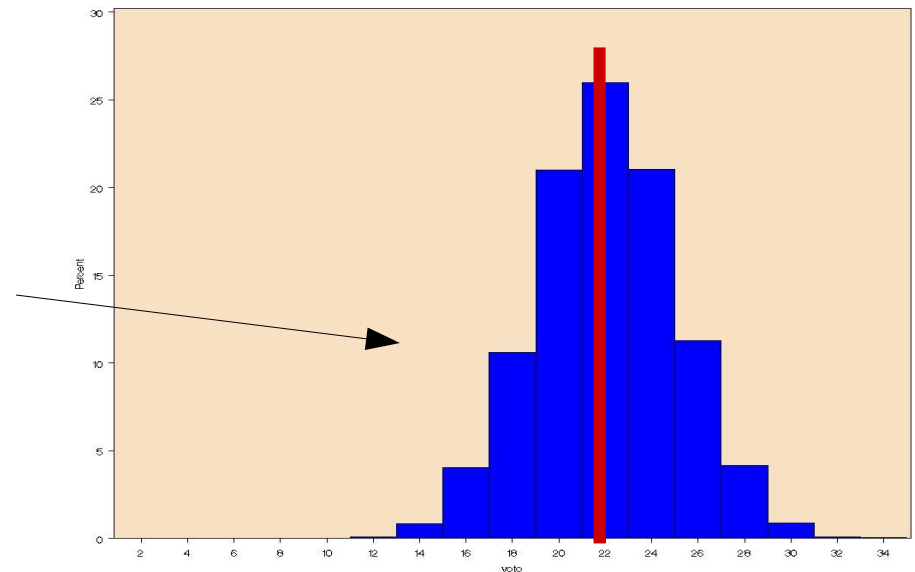


GLM Assumptions

$$y_i = a + e_i$$
$$\text{corr}(e_i, e_j) = 0$$

Random variations are
independent and normally
distributed

$$e_i \sim N(0, \sigma)$$



GLM

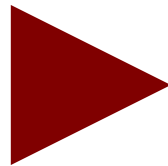
When the assumptions are NOT met because the data, and thus the errors, have more complex structures, we generalize the GLM to the Linear Mixed Model

The Linear Mixed Model is the statistical model underlying **multilevel models** and **repeated measures** analysis

Linear Mixed Model

GLM

Regression
T-test
ANOVA
ANCOVA
Moderation
Mediation
Path Analysis



LMM

Random coefficients models
Random intercept regression models
One-way ANOVA with random effects
One-way ANCOVA with random effects
Intercepts-and-slopes-as-outcomes models
Multi-level models

Example “beers”

Let's consider the case where the beer-smile research was conducted by gathering data in several different bars

bar

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	a	3	1.3	1.3	1.3
	b	14	6.0	6.0	7.3
	c	22	9.4	9.4	16.7
	d	21	9.0	9.0	25.6
	e	14	6.0	6.0	31.6
	f	20	8.5	8.5	40.2
	g	24	10.3	10.3	50.4
	h	12	5.1	5.1	55.6
	i	16	6.8	6.8	62.4
	l	22	9.4	9.4	71.8
	m	21	9.0	9.0	80.8
	n	15	6.4	6.4	87.2
	o	16	6.8	6.8	94.0
	p	11	4.7	4.7	98.7
	q	3	1.3	1.3	100.0
	Total	234	100.0	100.0	

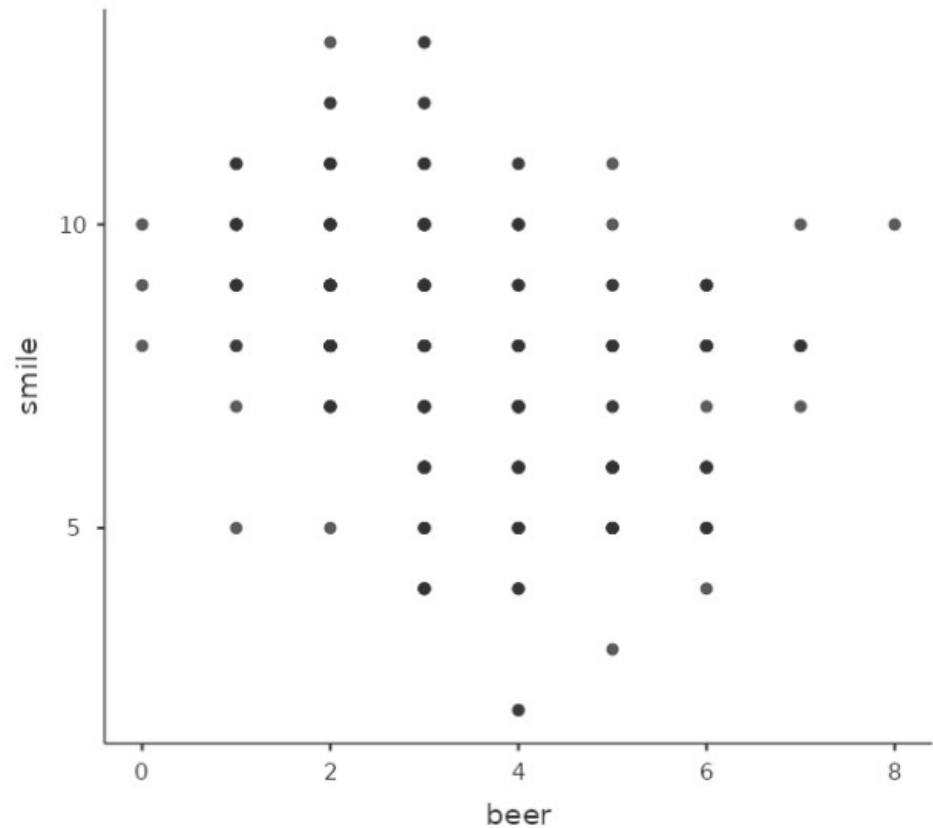
For each participant
we measured # of
beers and # of smiles

For a total of 234 participants

Example “beers” 2

As compared with the example with a few participants, now we have a very different scatterplot

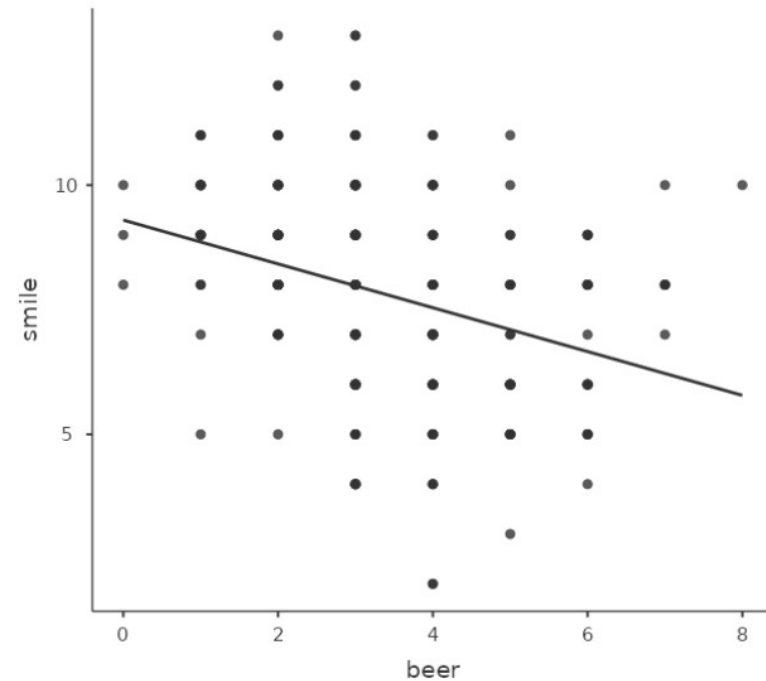
Scatterplot



Example “beers” 2

A simple regression confirms that results are indeed different

Scatterplot



Negative effect

Fixed Effects Parameter Estimates

Names	Estimate	SE	95% Confidence Interval		β	df	t	p
			Lower	Upper				
(Intercept)	7.765	0.130	7.508	8.022	0.000	232	59.503	< .001
beer	-0.440	0.085	-0.608	-0.271	-0.320	232	-5.147	< .001

Why

Results may be biased by a mis-specification of the model, where the structure of the data is not taken into account

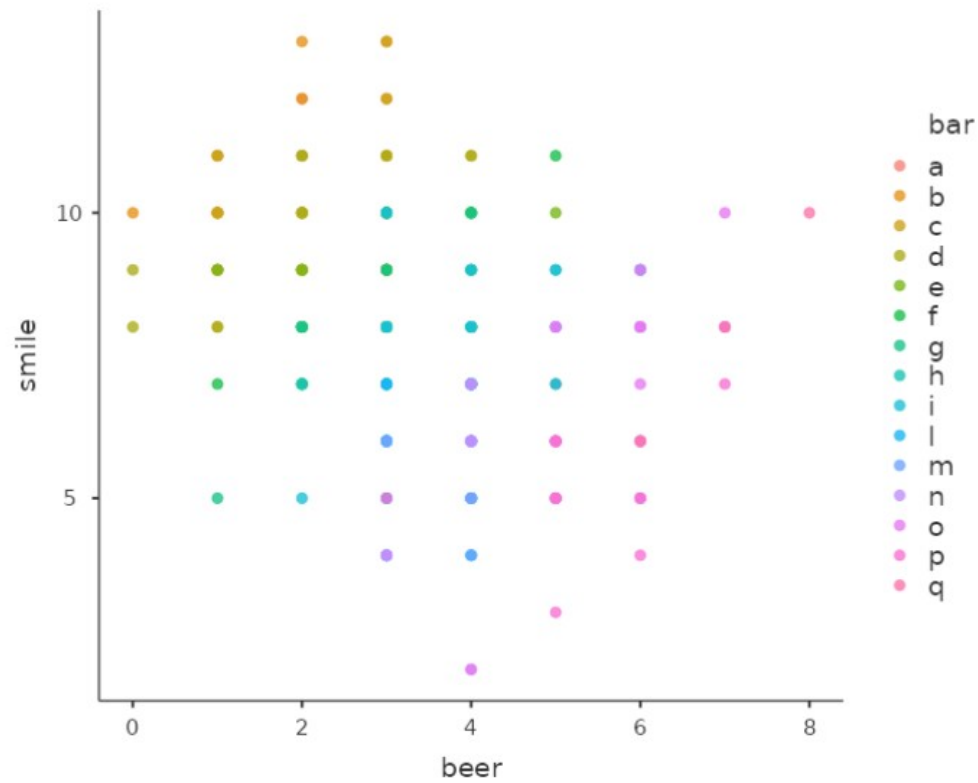
- In fact:
 - Subjects are sampled in clusters specified by **bars**
 - Each bar may have specific characteristics (quality, entertainment, etc) that may affect the measured variables
 - Subjects within the same bar may be more similar than across bars

Scatterplot by Bar

Let's see the data broken down by bar

Bar

Scatterplot

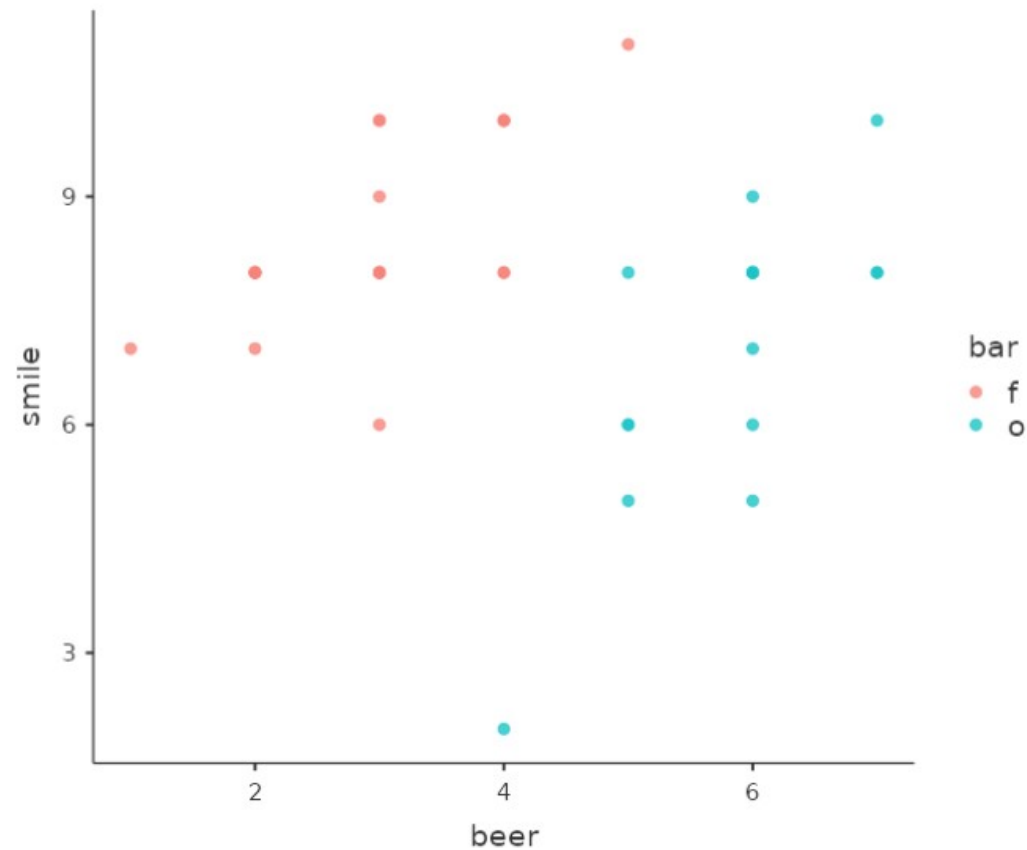


Scatterplot by Bar

Let's see the data only for bar “f” and “o”

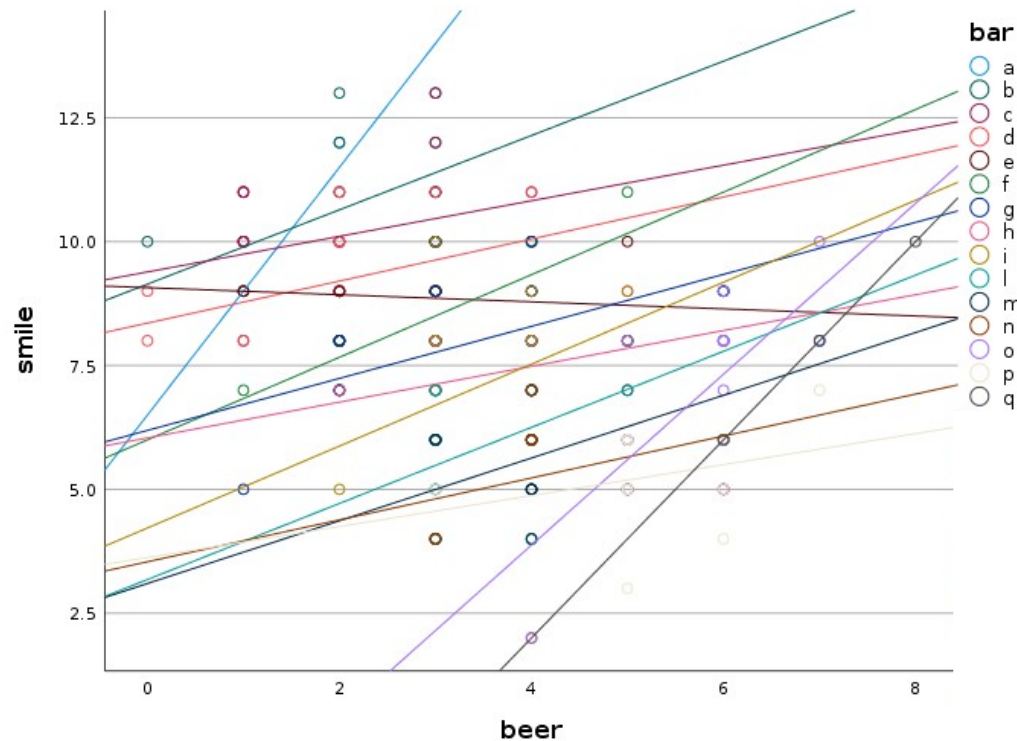
Bar

Scatterplot



Scatterplot by Bar

It seems that the relations between IV and DV is positive, but within each bar

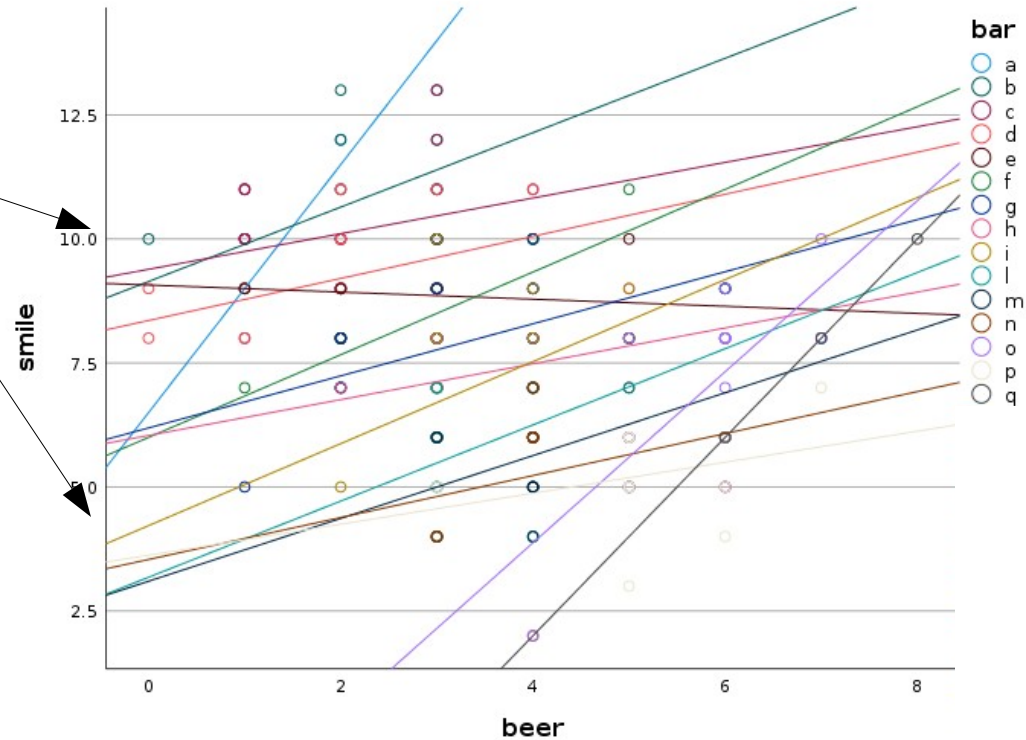


Bar

Scatterplot by Bar

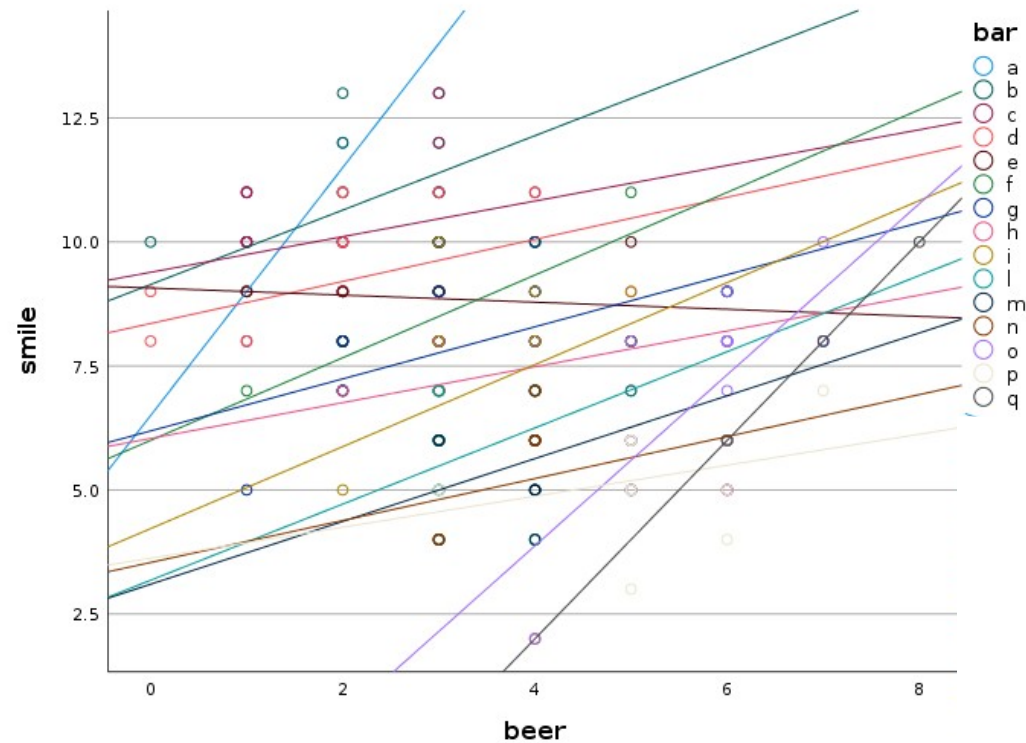
Bar

Intercepts seem to
be different from
bar to bar



Scatterplot by Bar

Bar



Slopes are all positive

Slopes seem to vary across bars

The Model

- It seems that considering the participants as all equivalent and independent one each other (GLM assumption) does not fit our data
- It seems that a better model should allow each bar (each cluster) to have a different regression line (a different intercept and **b** coefficient)

The Model

- Let's define a model with a regression line for each cluster

 y_{ij}

Smiles of subject i in the cluster j

$$\hat{y}_{ia} = a_a + b_a \cdot x_{ia}$$

$$\hat{y}_{ib} = a_b + b_b \cdot x_{ib}$$

$$\hat{y}_{ic} = a_c + b_c \cdot x_{ic}$$

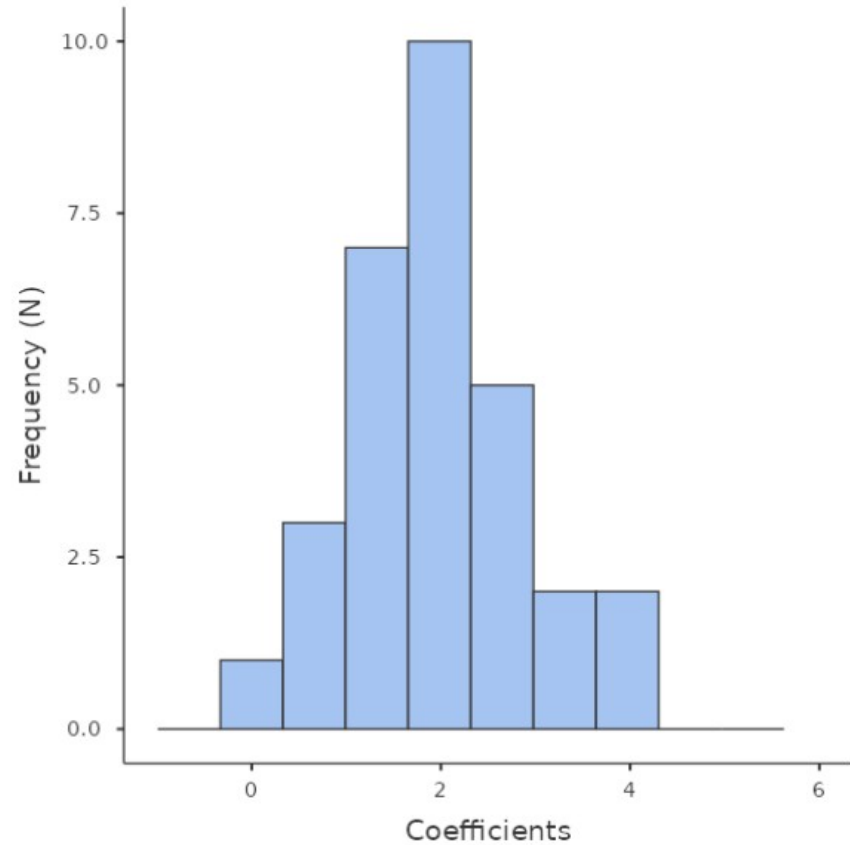
$$\hat{y}_{ij} = a_j + b_j \cdot x_{ij}$$

In these regressions the coefficients may vary from cluster to cluster: they are not **Fixed**

Varying coefficients

- If coefficients may vary, they will have a distribution

A possible distribution of coefficients b estimated for different clusters

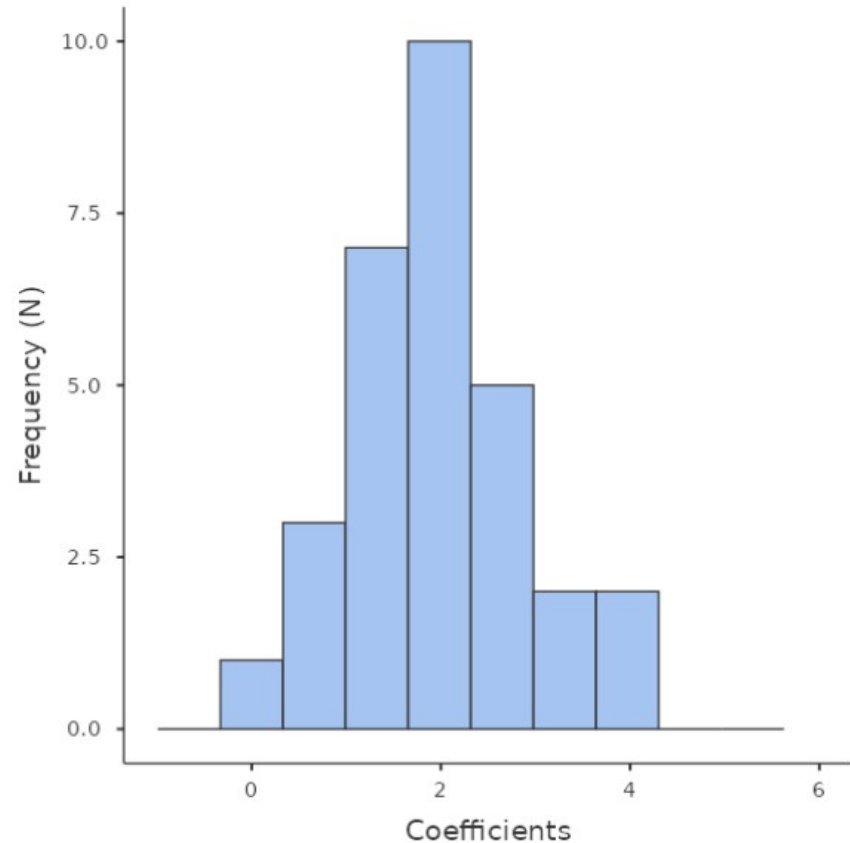


Random coefficients

- Varying coefficients are called **random coefficients**

Coefficients will exhibit variability: They are **random coefficients**

That is: in the **population** there exist different coefficients, a sample of which we estimated using the clustered data

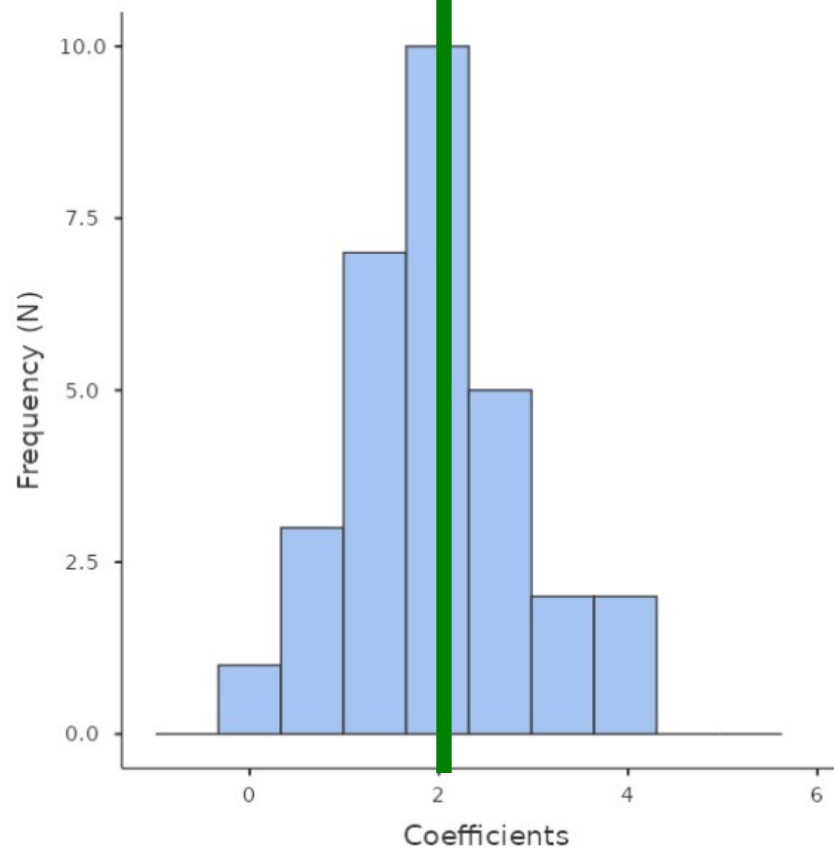


Average of the coefficients

- If coefficients vary as a variable in the population, they will have a mean and a variance, that we can estimate in our data

Mean

$$\bar{b} = \frac{\sum_j b_j}{k}$$



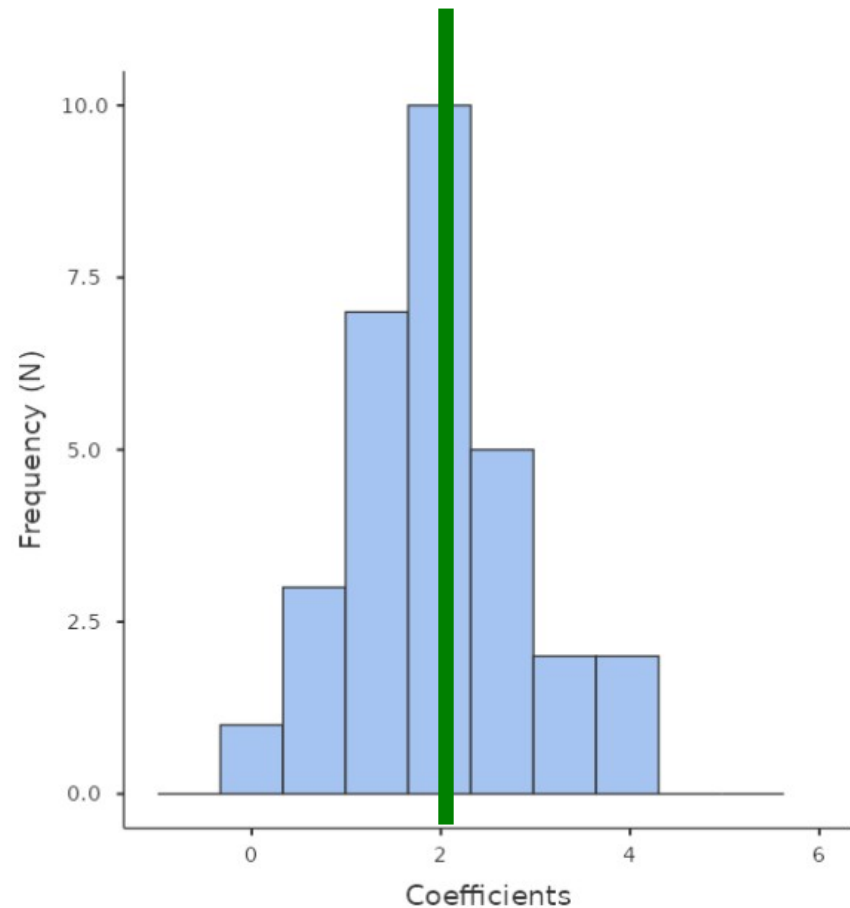
Fixed coefficients

- If coefficients vary as a variable in the population, they will have a mean and a variance, that we can estimate in our data

Mean

$$\bar{b} = \frac{\sum_j b_j}{k}$$

Recall the mean is a fixed parameter for a distribution, and so is the mean of the coefficients: it is a **fixed effect**



The Model

- We can now define a model with a regression for each cluster and the mean values of coefficients

One regression per cluster

$$\hat{y}_{ij} = a_j + b_j \cdot x_{ij}$$

Each coefficient is defined as the deviation from the mean coefficient

$$b'_j = b_j - \bar{b}$$

Overall model

$$\hat{y}_{ij} = a_j + b'_j \cdot x_{ij} + \bar{b} \cdot x_{ij}$$

The Model

- We can now define a model with a regression for each cluster and the mean value of coefficients

Overall model

$$\hat{y}_{ij} = a_j + b'_j \cdot x_{ij} + \bar{b} \cdot x_{ij}$$

**Random
coefficients**

Fixed coefficient

The mixed model

- The same goes for the intercepts

One regression per cluster

$$\hat{y}_{ij} = a_j + b_j \cdot x_{ij}$$

Intercepts as deviations from the average intercept

$$a'_j = a_j - \bar{a}$$

Overall model

$$\hat{y}_{ij} = \bar{a} + a'_j + \bar{b} \cdot x_{ij} + b'_j \cdot x_{ij}$$

The mixed model

- We can now define a model with a regression for each cluster and the mean values of coefficients

Overall model

$$\hat{y}_{ij} = \bar{a} + a'_j + \bar{b} \cdot x_{ij} + b'_j \cdot x_{ij}$$

**Random
coefficients**

Fixed coefficients

**A GLM which contains both fixed and
random effects is called a
Linear Mixed Model**

GLM as a special case

It is clear that everything we know for the GLM applies here: the GLM is in fact a special case of the LMM, where there are not random effects

LMM

$$\hat{y}_{ij} = \bar{a} + a'_j + \bar{b} \cdot x_{ij} + b'_j \cdot x_{ij}$$

GLM

$$\hat{y}_{ij} = \bar{a} + \bar{b} \cdot x_{ij}$$

The mixed model

- In practice, mixed models allow to estimate the kind of effects we can estimate with the GLM, but they allow the effects to vary across clusters.
- Effects that vary across clusters are called **random effects**
- Effects that do not vary (the ones that are the same across clusters) are said to be **fixed effects**

The mixed model

- To specify a correct model, we only need to understand if there are **clusters of cases** (measures or subjects) and decide which coefficients (intercepts or b coefficients) may vary across those clusters
- The fixed effects of the model are interpreted like in the GLM (regression/ANOVA)
- **Random effects** are generally not interpreted, but we can look at their variance to decide to keep them as random (variance >0) or fix them.
- In this way we take into the account the dependence among data

Building a model

To build a model in a simple way, we need to answer very few questions:

- What is (are) the cluster variable(s)?
- What are the fixed effects?
- What are the random effects?

A clustering variable

- **What is (are) the cluster variable(s)?**
- What are the fixed effects?
- What are the random effects?
 - Any variable that groups observations (cases or measurements) such that scores may be more similar within each group than across groups
 - Any variable whose levels (groups) are a sample of a larger population of levels (groups)
 - Example: bars created groups of scores (participants) that may be more similar within the bar than across bars

Fixed effects

- What is (are) the cluster variable(s)?
- **What are the fixed effects?**
- What are the random effects?
 - Any effect that we are interested in on average (as in a standard ANOVA/Regression)
 - Example: the effect of beer on smiles in general

Fixed effects

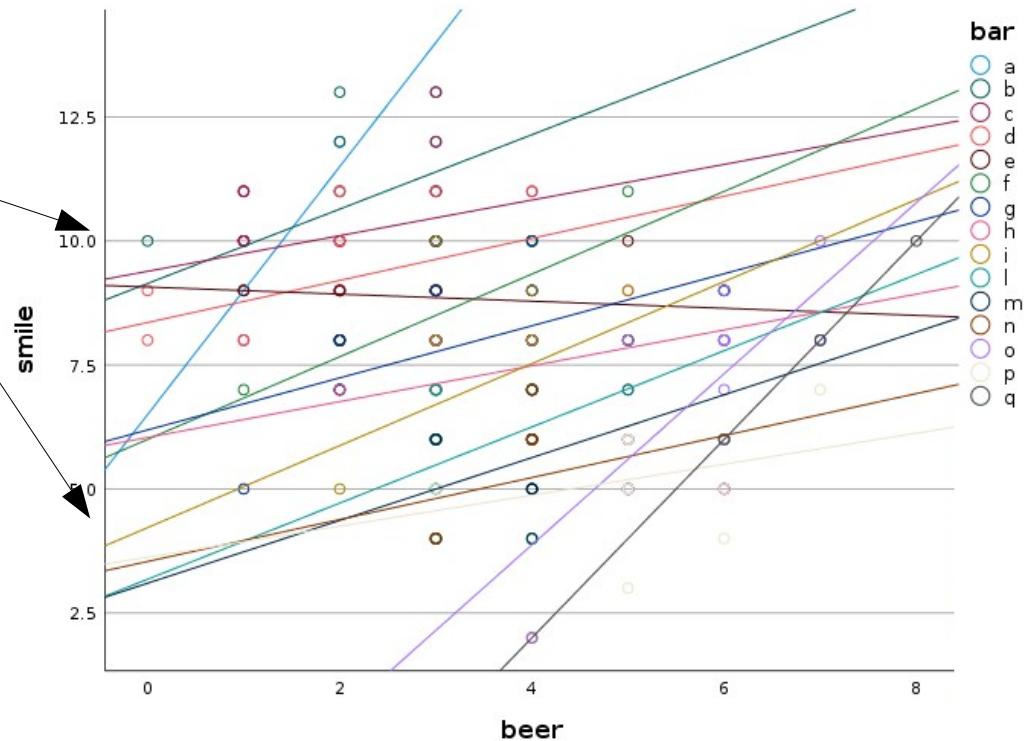
- What is (are) the cluster variable(s)?
- What are the fixed effects?
- **What are the random effects?**
 - Any effect that may vary from cluster to cluster
 - (Thus:) **Any effect that can be computed within each cluster**
 - Example: the intercepts and the effect of beer on smiles each bar

Beers at the bar

We start with a simple model

Bar

Intercepts seem to
be different from
bar to bar



Beers at the bar

We define a model where each cluster is allow to have a different intercept, the rest of the model is like a standard regression

$$y_{ij} = \bar{a} + a_j + \bar{b} \cdot x_{ij} + e_{ij}$$

- Fixed effects? Intercept and beer effect
- Random effects? Intercepts
- Clusters? bar

Authors and books may call this model:
Random-intercepts regression
or
Intercepts-as-outcomes model

Beers at the bar

We define a model where each cluster is allow to have a different intercept, the rest of the model is like a standard regression

$$y_{ij} = \bar{a} + a_j + \bar{b} \cdot x_{ij} + e_{ij}$$

- Fixed effects? Intercept and beer effect
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- Clusters? bar

Authors and books may call this model:

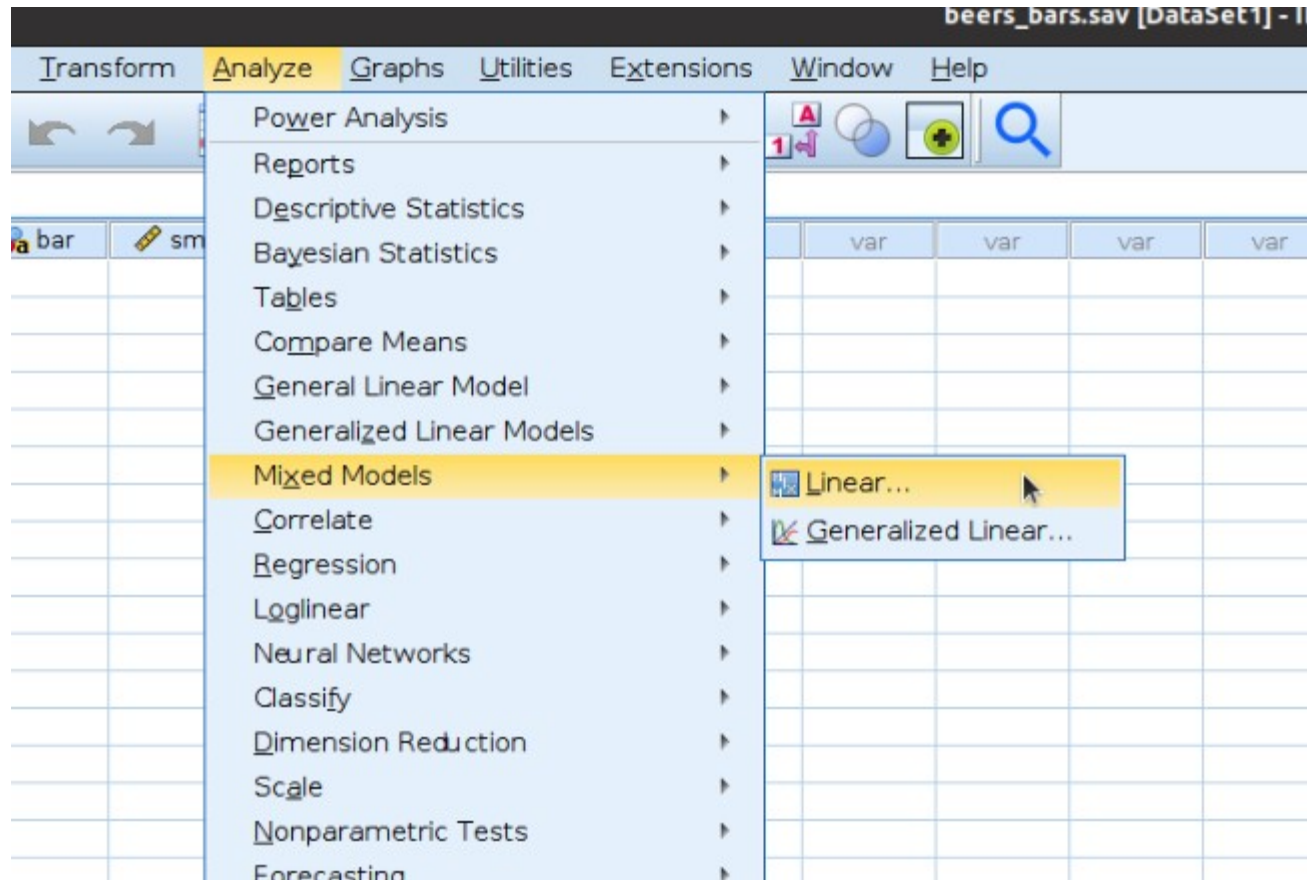
Random-intercepts regression

or

Intercepts-as-outcomes model

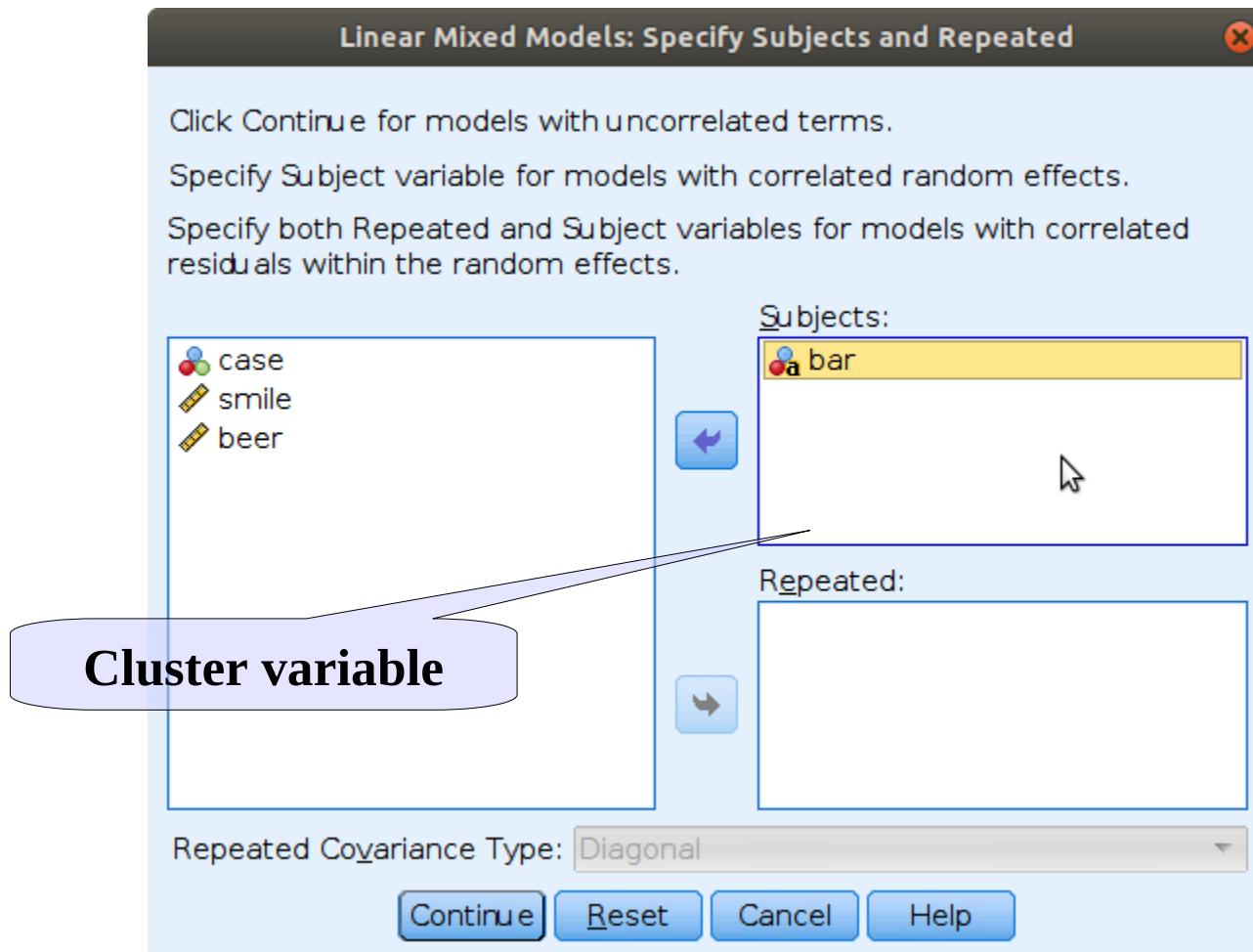
SPSS Input

Analyze → Mixed Models → Linear



SPSS Input

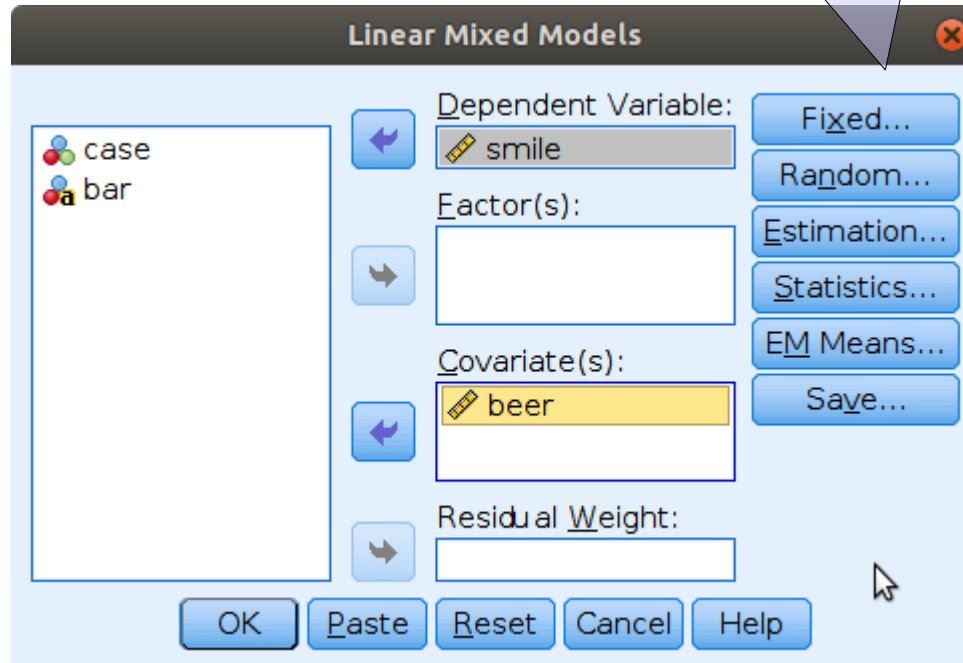
Analyze → Mixed Models → Linear



SPSS Input

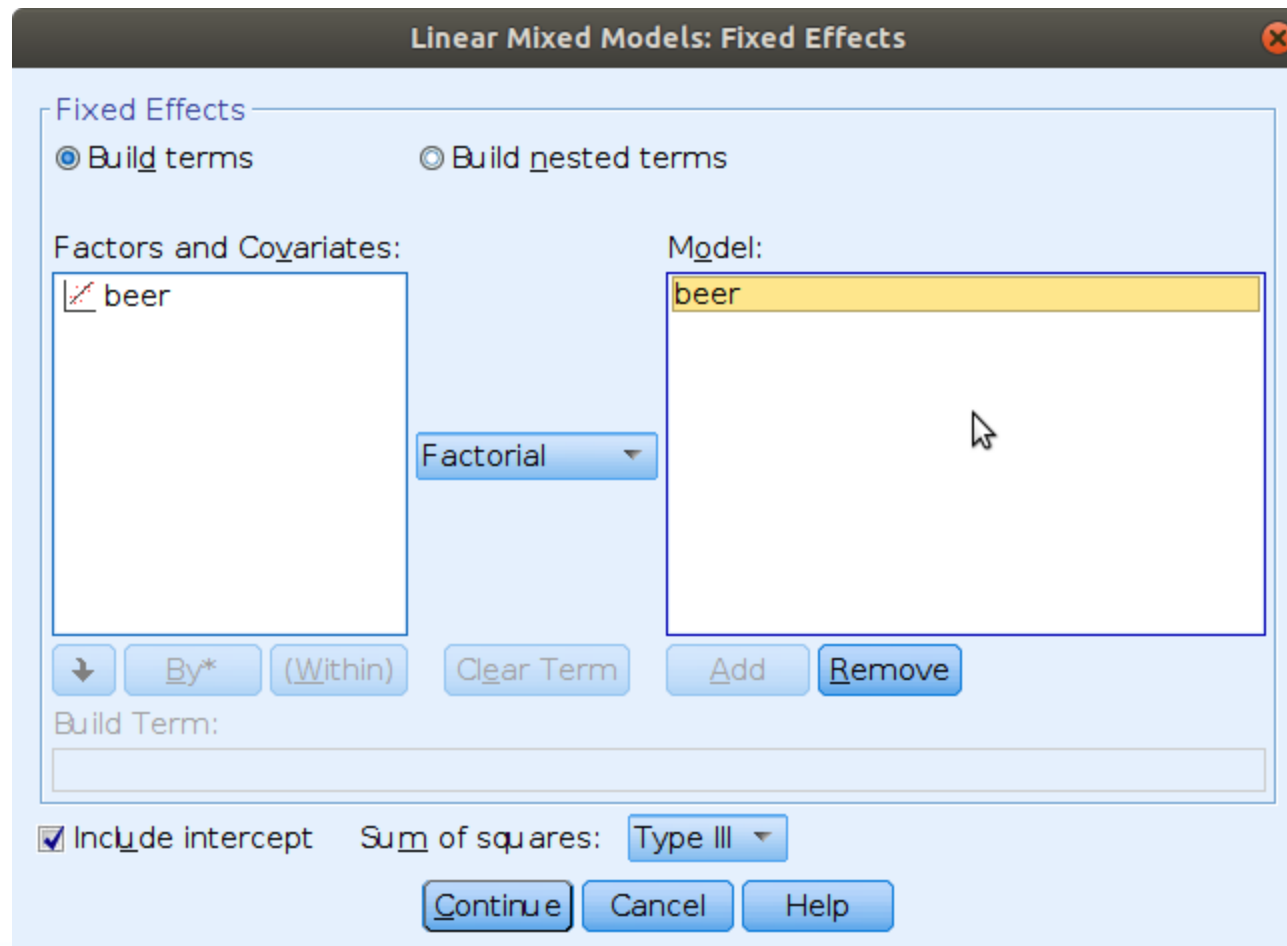
Analyze → Mixed Models → Linear

Then select Fixed



SPSS Input

Analyze → Mixed Models → Linear



The image shows the 'Linear Mixed Models: Fixed Effects' dialog box in SPSS. The 'Fixed Effects' section has two radio buttons: 'Build terms' (selected) and 'Build nested terms'. Below these are two list boxes: 'Factors and Covariates:' containing 'beer' and 'Model:' containing 'beer'. A 'Factorial' dropdown menu is positioned between the two list boxes. At the bottom of the dialog, there is a 'Build Term:' text box, a checked 'Include intercept' checkbox, a 'Sum of squares:' dropdown set to 'Type III', and three buttons: 'Continue', 'Cancel', and 'Help'.

Linear Mixed Models: Fixed Effects

Fixed Effects

☒ Build terms ☐ Build nested terms

Factors and Covariates:

beer

Model:

beer

Factorial

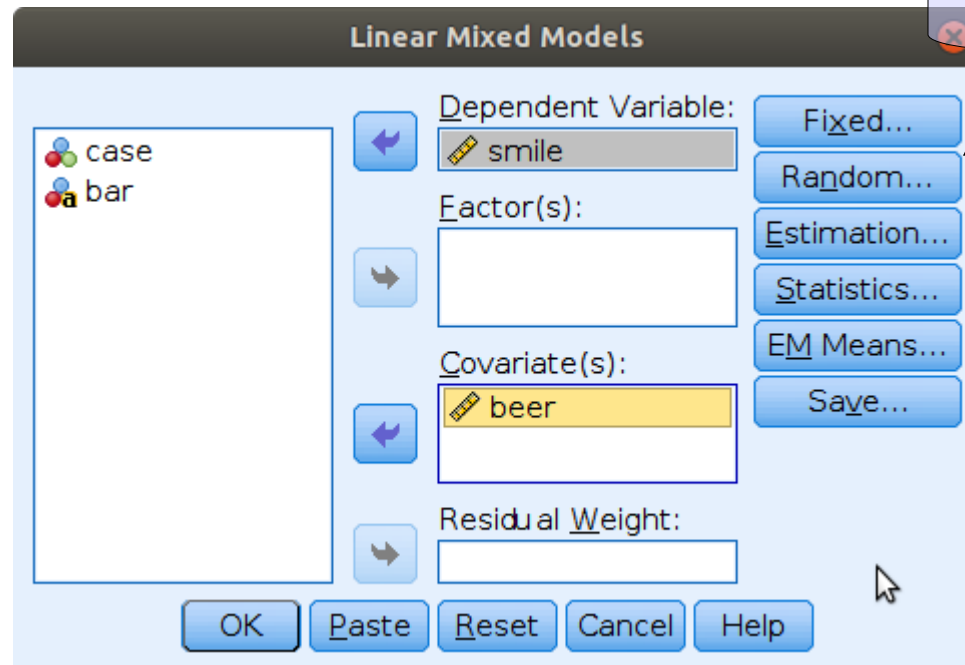
Build Term:

☒ Include intercept Sum of squares: Type III

Continue Cancel Help

SPSS Input

Analyze → Mixed Models → Linear



**Then select
Random**

SPSS Input

Analyze → Mixed Models → Linear

Linear Mixed Models: Random Effects

Random Effect 1 of 1

Previous Next

Covariance Type: Variance Components

Random Effects

☒ Build terms ☐ Build nested terms ☒ Include intercept

Factors and Covariates: Model:

beer

Factorial

↓ By* (Within) Clear Term Add Remove

Build Term:

Subject Groupings

Subjects: Combinations:

a bar

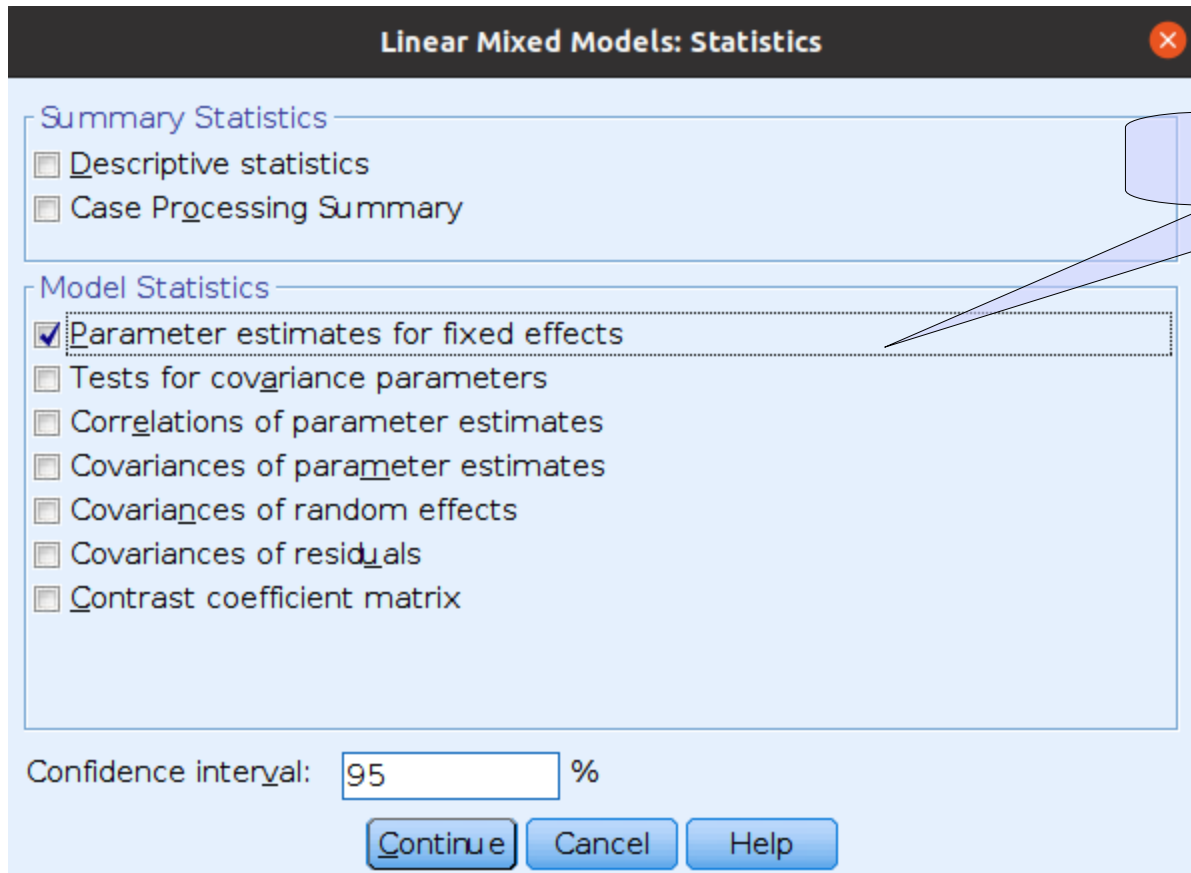
↺

Continue Cancel Help

Random Intercept

SPSS Input

Analyze → Mixed Models → Linear



SPSS syntax

$$y_{ij} = \bar{a} + a_j + \bar{b} \cdot x_{ij} + e_{ij}$$

MIXED smile WITH beer by bar

/CRITERIA=CIN(95) MXITER(100) MXSTEP(10) SCORING(1) SINGULAR(0.000000000001) HCONVERGE(0,
ABSOLUTE) LCONVERGE(0, ABSOLUTE) PCONVERGE(0.000001, ABSOLUTE)

/FIXED=beer| SSTYPE(3)

/METHOD=REML

/print solution TESTCOV

/random intercept | SUBJECT(bar) COVTYPE(un).

Fixed effects (intercept is included by default)

Random effects

Cluster variable

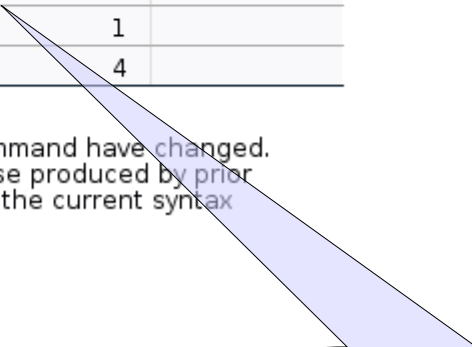
SPSS Output

Let's see if the model is how intended

Model Dimension ^a					
		Number of Levels	Covariance Structure	Number of Parameters	Subject Variables
Fixed Effects	Intercept	1		1	
	beer	1		1	
Random Effects	Intercept ^b	1	Variance Components	1	bar
Residual				1	
Total		3		4	

a. Dependent Variable: smile.

b. As of version 11.5, the syntax rules for the RANDOM subcommand have changed. Your command syntax may yield results that differ from those produced by prior versions. If you are using version 11 syntax, please consult the current syntax reference guide for more information.



OK!

SPSS Output

We then check the variability of the random effects. If there is variability across bars, it means we were right to model the coefficients as random

Covariance Parameters

Estimates of Covariance Parameters^a

Parameter	Estimate	Std. Error
Residual	1.451189	.139257
Intercept [subject = bar] Variance	5.816514	2.320621

a. Dependent Variable: smile.

Variance greater than 0

SPSS Output

If everything is fine, we interpret the fixed effects as in any other GLM (regression)

Estimates of Fixed Effects^a

Parameter	Estimate	Std. Error	df	t	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
Intercept	5.841778	.695573	19.205	8.399	<.001	4.386978	7.296578
beer	.552973	.080740	229.072	6.849	<.001	.393884	.712061

a. Dependent Variable: smile.

Intercept: On average, for zero beers we expect 5.8 smiles

SPSS Output

If everything is fine, we interpret the fixed effects as in any other GLM (regression)

Estimates of Fixed Effects^a

Parameter	Estimate	Std. Error	df	t	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
Intercept	5.841778	.695573	19.205	8.399	<.001	4.386978	7.296578
beer	.552973	.080740	229.072	6.849	<.001	.393884	.712061

a. Dependent Variable: smile.

Intercept: On average, as beers increase on 1 unit, we expect smile to increase of .552 smiles

SPSS Output

We also get the overall omnibus tests

Fixed Effects

Type III Tests of Fixed Effects^a

Source	Numerator df	Denominator df	F	Sig.
Intercept	1	19.205	70.535	<.001
beer	1	229.072	46.906	<.001

a. Dependent Variable: smile.

**These are equivalent to the GLM
F-tests**

Beers at the bar 2

We can now try a model where also the **b** coefficients are allow to vary across clusters

$$y_{ij} = \bar{a} + a_j + \bar{b} \cdot x_{ij} + b \cdot x_{ij} + e_{ij}$$

- Fixed effects? Intercept and beer effect
- Random effects? Intercepts and b coefficients
- Clusters? bar

Some authors may call this model:
Random-coefficients regression
or
Intercepts- and Slopes-as-outcomes model

Jamovi

- We keep the same setup for the variables

Mixed Model

Dependent Variable

→ smile

Factors

→

Covariates

→ beer

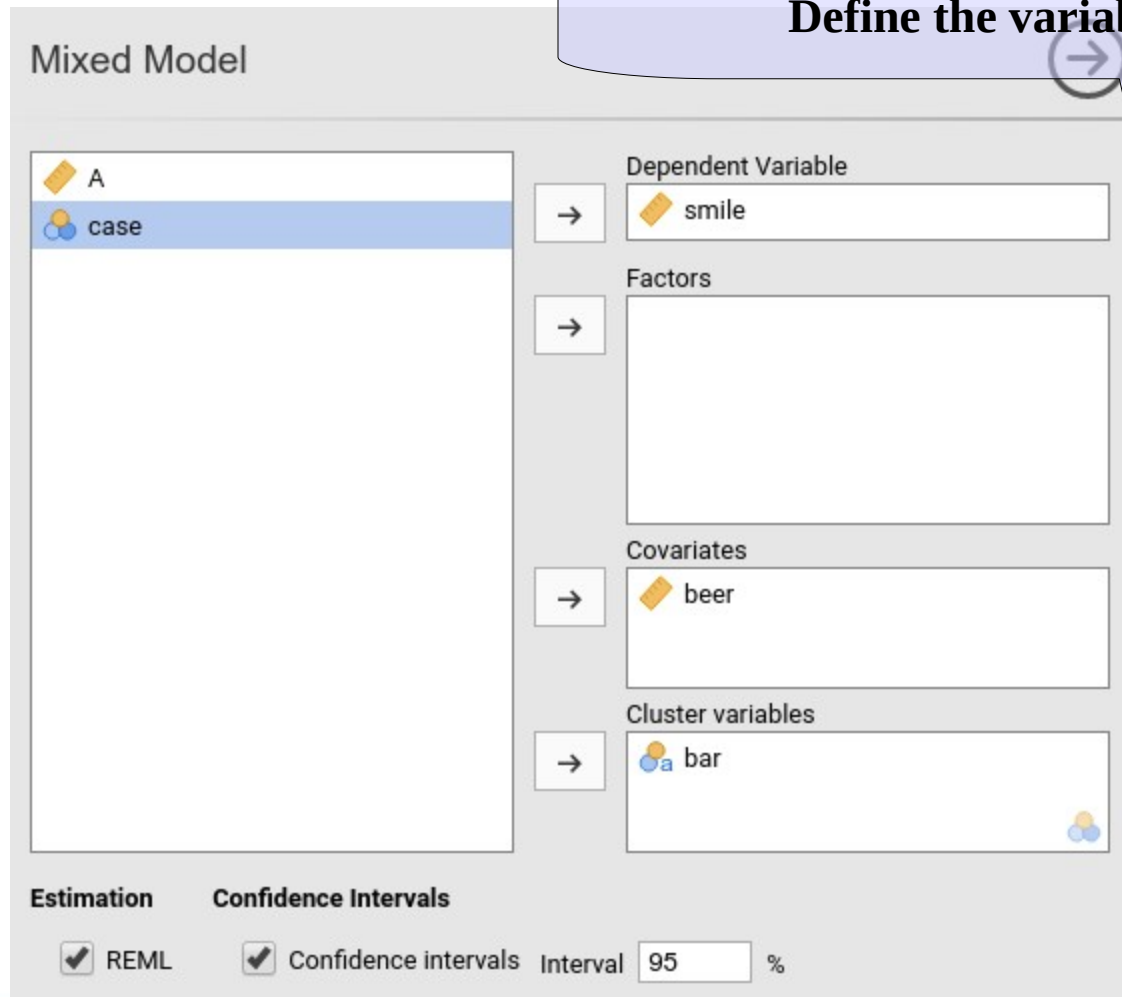
Cluster variables

→ bar

Estimation

Confidence Intervals

☒ REML ☒ Confidence intervals Interval 95 %



Define the variables role

- We add the effect of beer as a random coefficient

Define the random component

The screenshot shows the 'Random Effects' dialog box in Jamovi. The 'Components' list on the left is empty. The 'Random Coefficients' list on the right contains 'Intercept | bar' and 'beer | bar'. Below the lists, the 'Effects correlation' section has three radio buttons: 'Correlated' (selected), 'Not correlated', and 'Correlated by block'. The 'Tests' section has a checkbox for 'LRT for Random Effects' which is unchecked.

Random Effects

Components

Random Coefficients

Intercept | bar
beer | bar

Effects correlation

Correlated
Not correlated
Correlated by block

Tests

LRT for Random Effects

- As soon as you define the random component, you get the results

Model Results

Model Fit

Type	R ²	df	LRT X ²	p
Conditional	0.822	4	203.003	< .001
Marginal	0.090	1	17.016	< .001

[4]

>

R-squared Marginal: How much variance can the fixed effects alone explain of the overall variance

R-squared Conditional: How much variance can the fixed and random effects together explain of the overall variance

- As soon as you define the random component, you get the results

Model Results

F-test for the main effect of beer

Fixed Effect Omnibus tests

	F	Num df	Den df	p
beer	36.057	1	7.234	< .001

Note. Satterthwaite method for degrees of freedom

- As soon as you define the random component, you get the results

**coefficients for the main effect of
beer**

Fixed Effects Parameter Estimates

Names	Estimate	SE	95% Confidence Interval		df	t	p
			Lower	Upper			
(Intercept)	7.610	0.633	6.368	8.851	12.928	12.013	< .001
beer	0.555	0.093	0.374	0.737	7.234	6.005	< .001

- As soon as you define the random component, you get the results

Random Components

Groups	Name	SD	Variance	ICC
bar	(Intercept)	2.417	5.842	0.803
	beer	0.167	0.028	
Residual		1.196	1.431	

Note. Number of Obs: 234 , groups: bar 15

Random Parameters correlations

Groups	Param.1	Param.2	Corr.
bar	(Intercept)	beer	-0.766

**Random coefficients
variances**

**Random coefficients
correlation**

Jamovi

- Jamovi can plot up to a 3-way interaction

Plots

Horizontal axis
→ beer

Separate lines
→

Separate plots
→

Display

☒ None
☐ Confidence intervals
Interval 95 %
☐ Standard Error

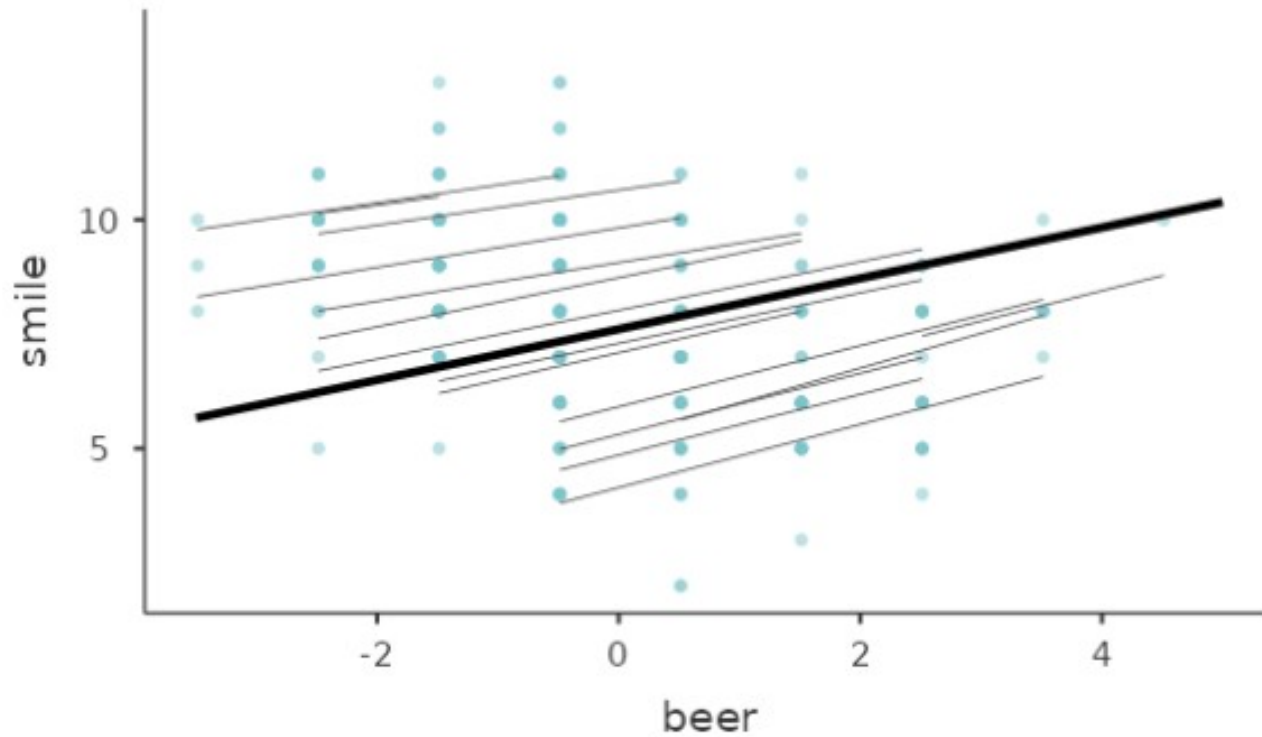
Plot

☒ Observed scores
☐ Y-axis observed range
☒ Random effects

Plot

- In this case is fixed and random effects regression lines

Effects Plots



Mixed Linear Models

- With the mixed model one can take into the account dependency among measures (within clusters) almost in any situation
- It allows applying the GLM logic to a broader range of designs
- Any kind of independent variables
- Generalizes to the generalized linear model (logistic etc)
- Efficient handling of missing values
- **Multi-level research designs**
- **Repeated measures designs**

Multi-level models

Multi-level models

- The Multi-level model is **not** a “statistical technique”!
- The Multi-level model is **an approach** to analyze multi-level designs
- **The multi-level model is estimated using a mixed model**
- What is peculiar:
 - The importance of the clustering variables (higher levels)
 - The research questions
 - The cluster level is called group level (*group=cluster in this terminology*)

Example

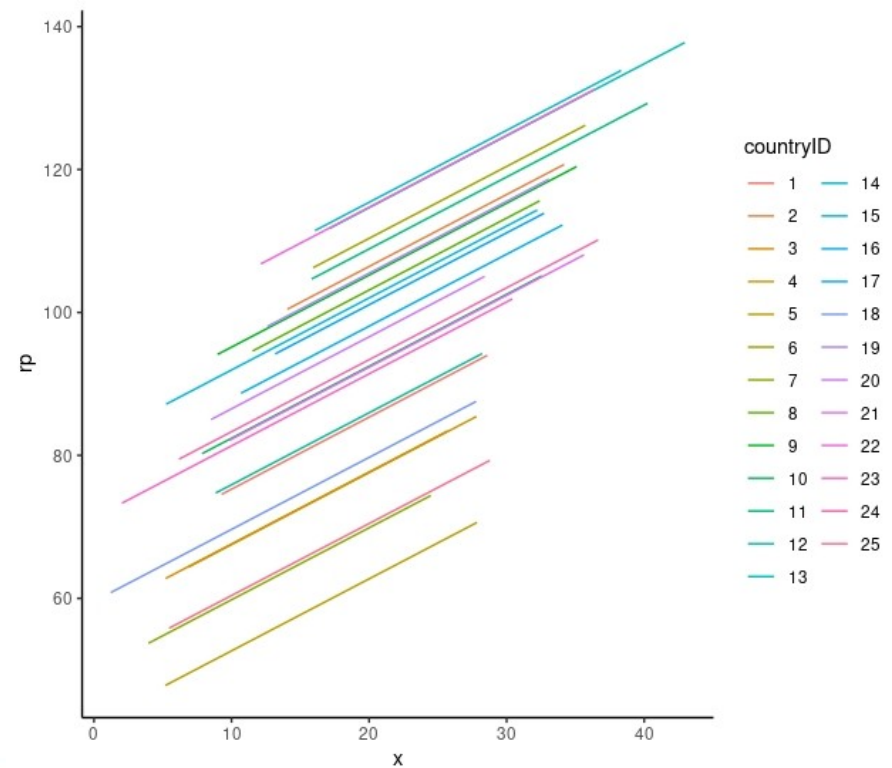
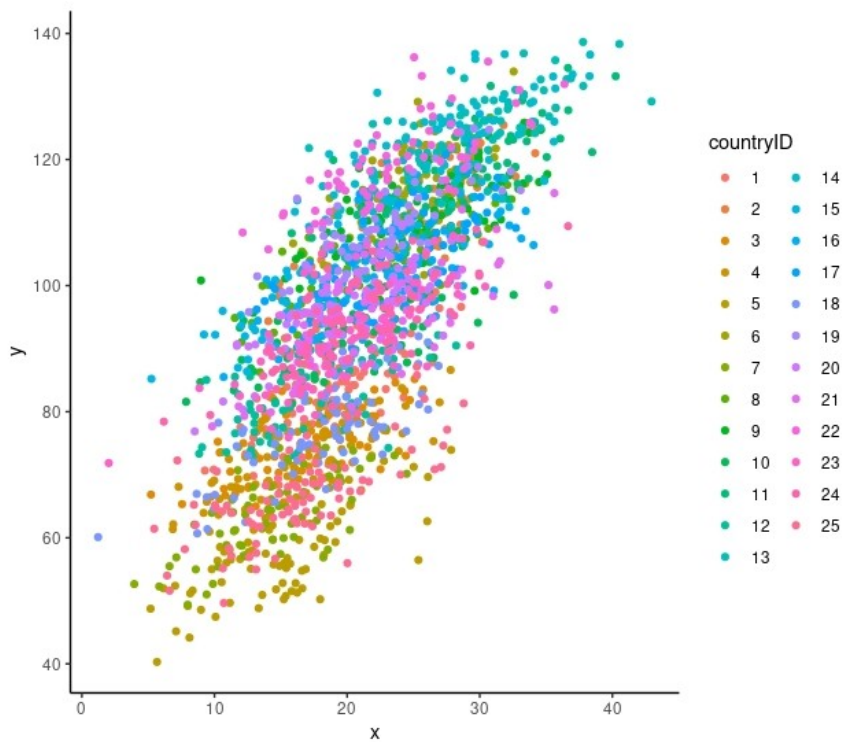
- Assume we have a multi-country research, in which we measured individuals (people) *charity contribution* and their *income (individual level)*.
- We can have information about country taxes regulations (*country level*)
- We are interested on the relationship between *contribution* and *income* at the **individual level and at the country level**

Questions

- We are interested on the relationship between cooperation and income at the **individual levels and at the country level**
- Independently of the country: Does people with higher income contribute more? (*individual level effect*)
- Independently of the people: Do countries with average higher income show higher average contribution? (*country level effect*)

Structure vs aim

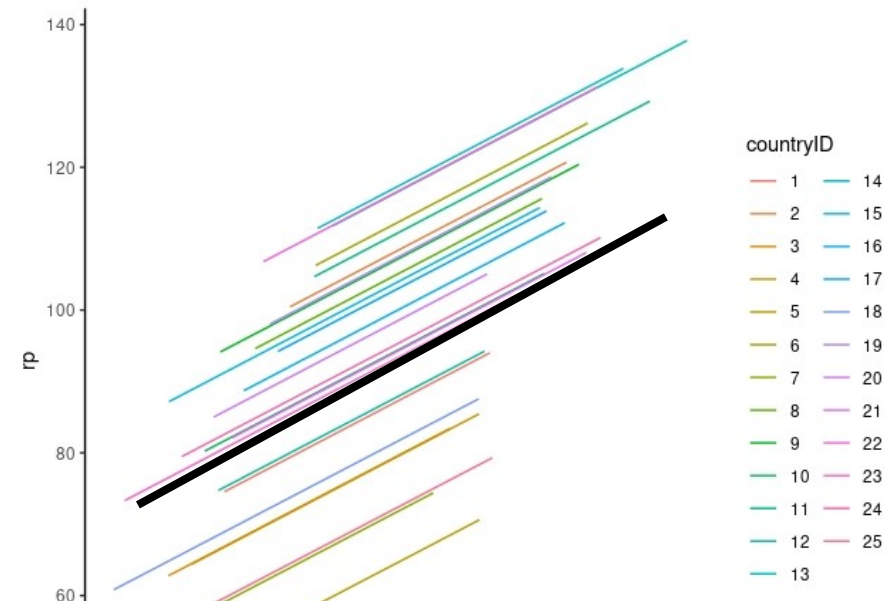
- If we only look at the data structure, we are in the beers at bars example
- But the research aim is different



Individual level

- If we fit a model like *beer at bars*, we only get the individual level effect, averaged across countries

Independently of the country: Does people with higher income contribute more? (*individual level effect*)



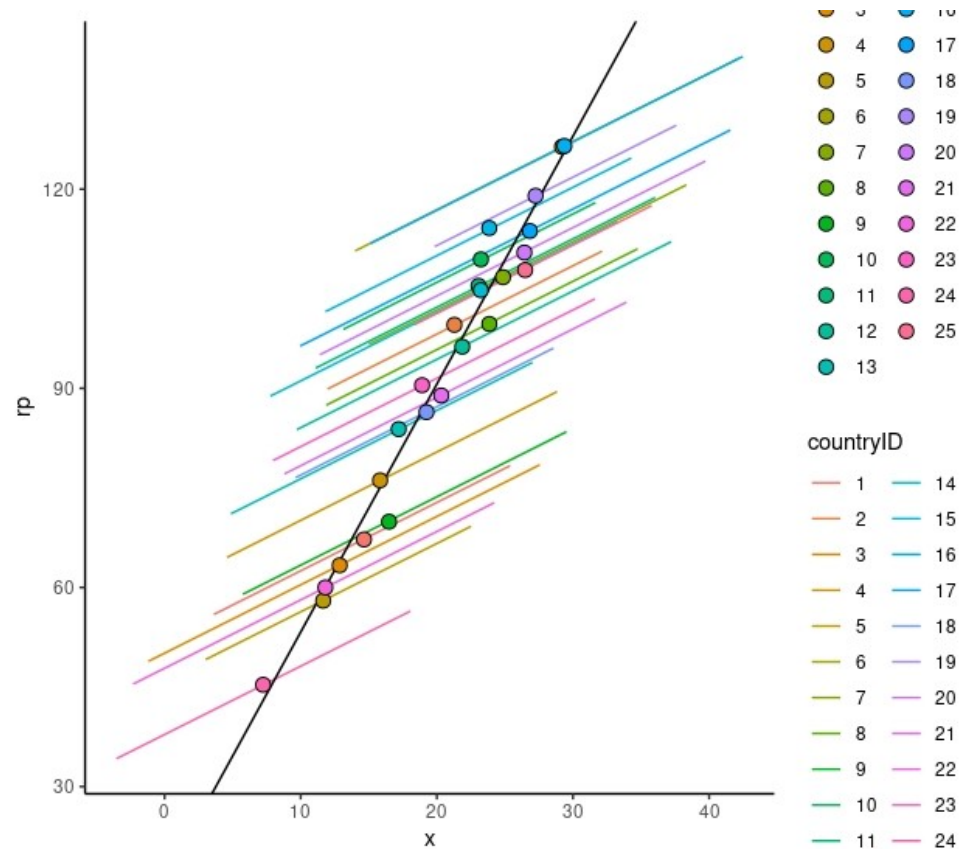
Fixed Effects Parameter Estimates

Names	Estimate	SE	95% Confidence Interval		df	t	p
			Lower	Upper			
(Intercept)	95.73	3.0129	89.830	101.64	24.0	31.8	< .001
x	1.01	0.0235	0.964	1.06	1928.6	43.0	< .001

Country level

- But income may have an effect also at the country (second) level

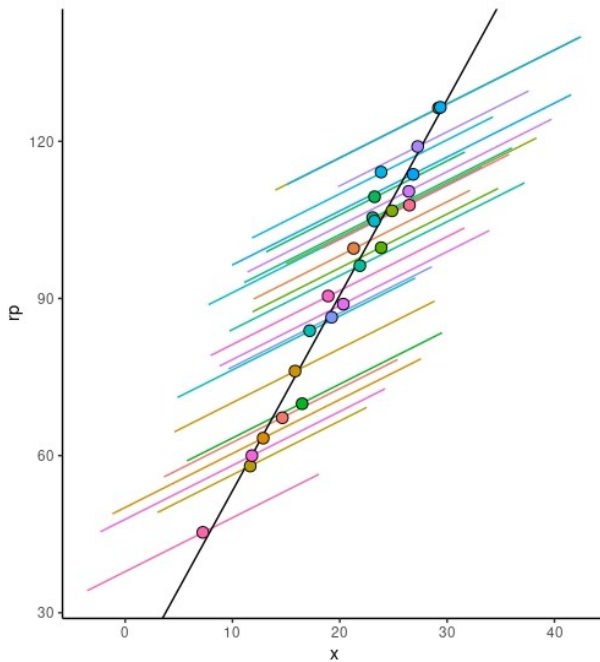
Independently of the people: Do countries with average higher income show higher average contribution? (*country level effect*)



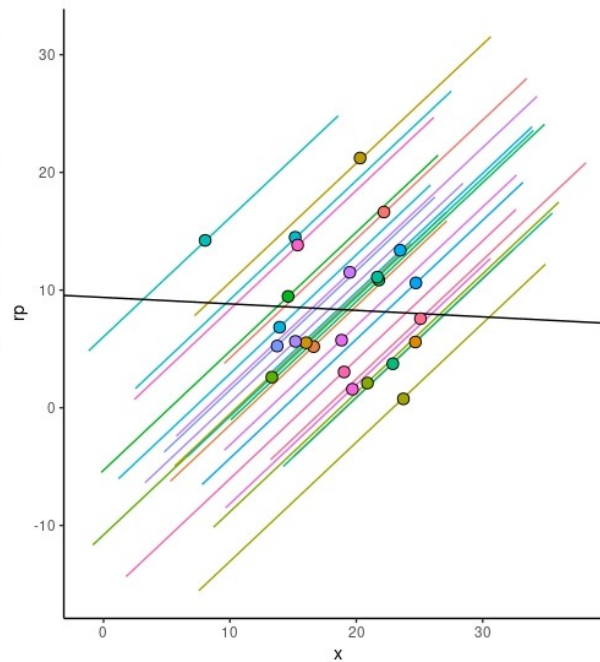
Country vs individual level

- The effect of a variable at each level is **independent** of the effect at any other level

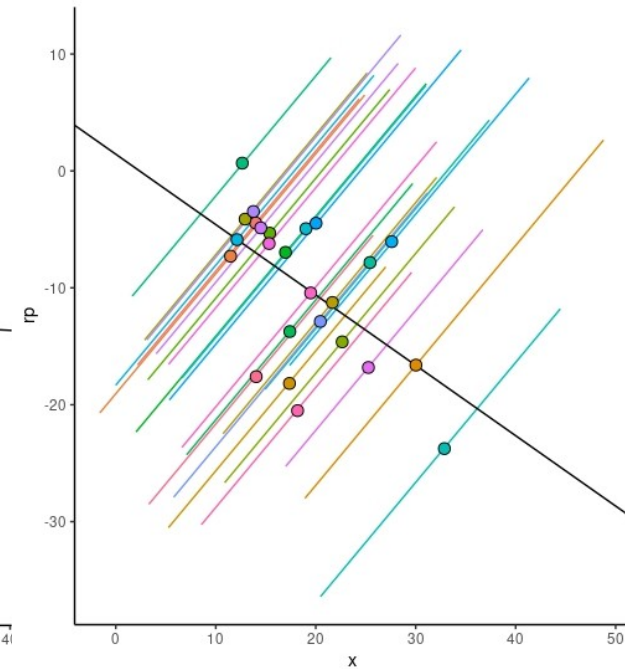
Individual +, country +



individual +, country 0



individual +, country -

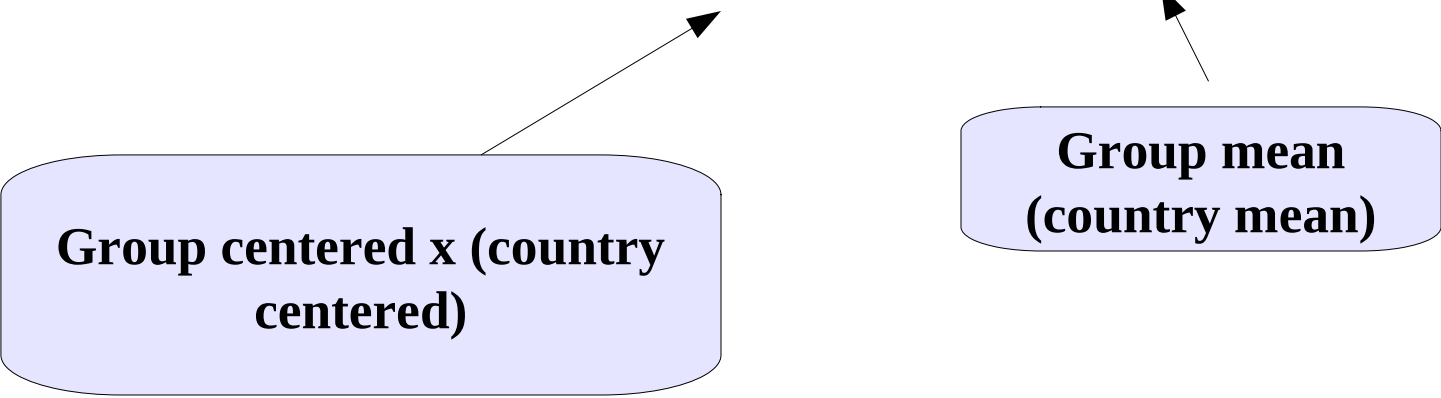


The mixed model

- To capture the effect of countries (second level) we should include the country levels means
- To make it independent of people levels, we group-center individual level x

$$\hat{y}_{ij} = \bar{a} + a'_j + b_1 \cdot (x_{ij} - \bar{x}_j) + b_2 \cdot \bar{x}_j$$

Group centered x (country centered)



**Group mean
(country mean)**

Coefficients

- The model returns the effects at level 1 (individuals) and level 2 (country)

$$\hat{y}_{ij} = \bar{a} + a'_j + b_1 \cdot (x_{ij} - \bar{x}_j) + b_2 \cdot \bar{x}_j$$

Independently of the country: Does people with higher income contribute more?

Do countries with average higher income show higher average contribution?

Data

- We simply compute two new variables: the group centered x and the group means

$$xcen = (x_{ij} - \bar{x}_j)$$

$$xm = \bar{x}_j$$

countryID	x	y	xm	xcen
10	12.919	88.400	20.547	-7.628
10	27.128	98.083	20.547	6.581
10	21.709	99.612	20.547	1.163
10	20.586	94.523	20.547	0.040
10	14.580	81.559	20.547	-5.967
10	28.697	106.248	20.547	8.150
10	8.937	84.683	20.547	-11.609
10	17.241	94.658	20.547	-3.306
10	18.252	84.677	20.547	-2.295
10	18.913	90.324	20.547	-1.633
11	33.202	129.347	28.333	4.870
11	23.598	114.079	28.333	-4.735
11	23.746	109.661	28.333	-4.587
11	26.870	113.275	28.333	-1.463
11	25.305	116.171	28.333	-3.028
11	31.789	114.460	28.333	3.456
11	18.956	119.912	28.333	-9.377
11	32.524	123.181	28.333	4.191
11	27.646	111.706	28.333	-0.717

Model

- We use them as independent variables

Mixed Model 

 A
 x
 rp



→

Dependent Variable
 y

→

Factors

→

Covariates
 xcen
 xm

→

Cluster variables
 countryID

Random Coefficients

Intercept | countryID
xcen | countryID

Results (fixed effect)

- And interpret the coefficients accordingly

Fixed Effects Parameter Estimates

Names	Estimate	SE	95% Confidence Interval		df	t	p
			Lower	Upper			
(Intercept)	95.65	1.3013	93.104	98.21	23.1	73.5	< .001
xcen	1.01	0.0278	0.954	1.06	21.8	36.3	< .001
xm	4.37	0.3126	3.757	4.98	23.1	14.0	< .001

Within each country: as people income increases of 1 unit, contribution increases of **1.01**

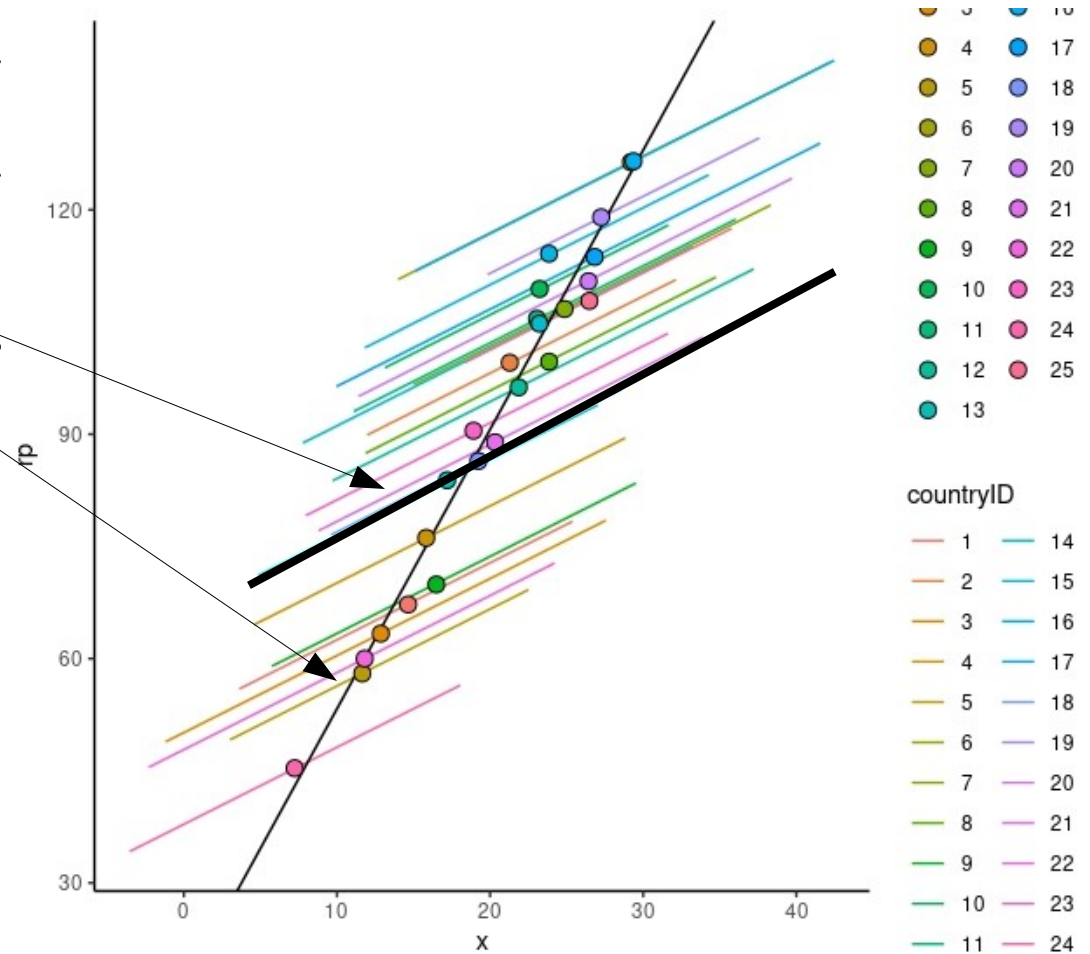
Across countries: As the average income of a country increases 1 unit, the average contribution increases of **4.37** units

Results (fixed effect)

- And interpret the coefficients accordingly

Fixed Effects Parameter Estimates

Names	Estimate	SE
(Intercept)	95.65	1.3013
xcen	1.01	0.0278
xm	4.37	0.3126



Multi-level models

- **The multi-level model is estimated using a mixed model**
- What is peculiar:
 - We want to estimate predictors effects at each level
 - We want to estimate higher level effect over and beyond lower level effect (contextual effects)

Mixed Linear Models

- With the mixed model one can take into the account dependency among measures (within clusters) almost in any situation
- It allows applying the GLM logic to a broader range of designs
- Any kind of independent variables
- Generalizes to the generalized linear model (logistic etc)
- Efficient handling of missing values
- Multi-level research designs
- **Repeated measures designs**

Repeated Measures Anova as a linear mixed model

Repeated Measures Anova as a linear mixed model

A repeated measures design

- Consider now a classical repeated measures design (within-subjects) the levels of the WS IV (5 different conditions) are represented by different measures taken on the same person

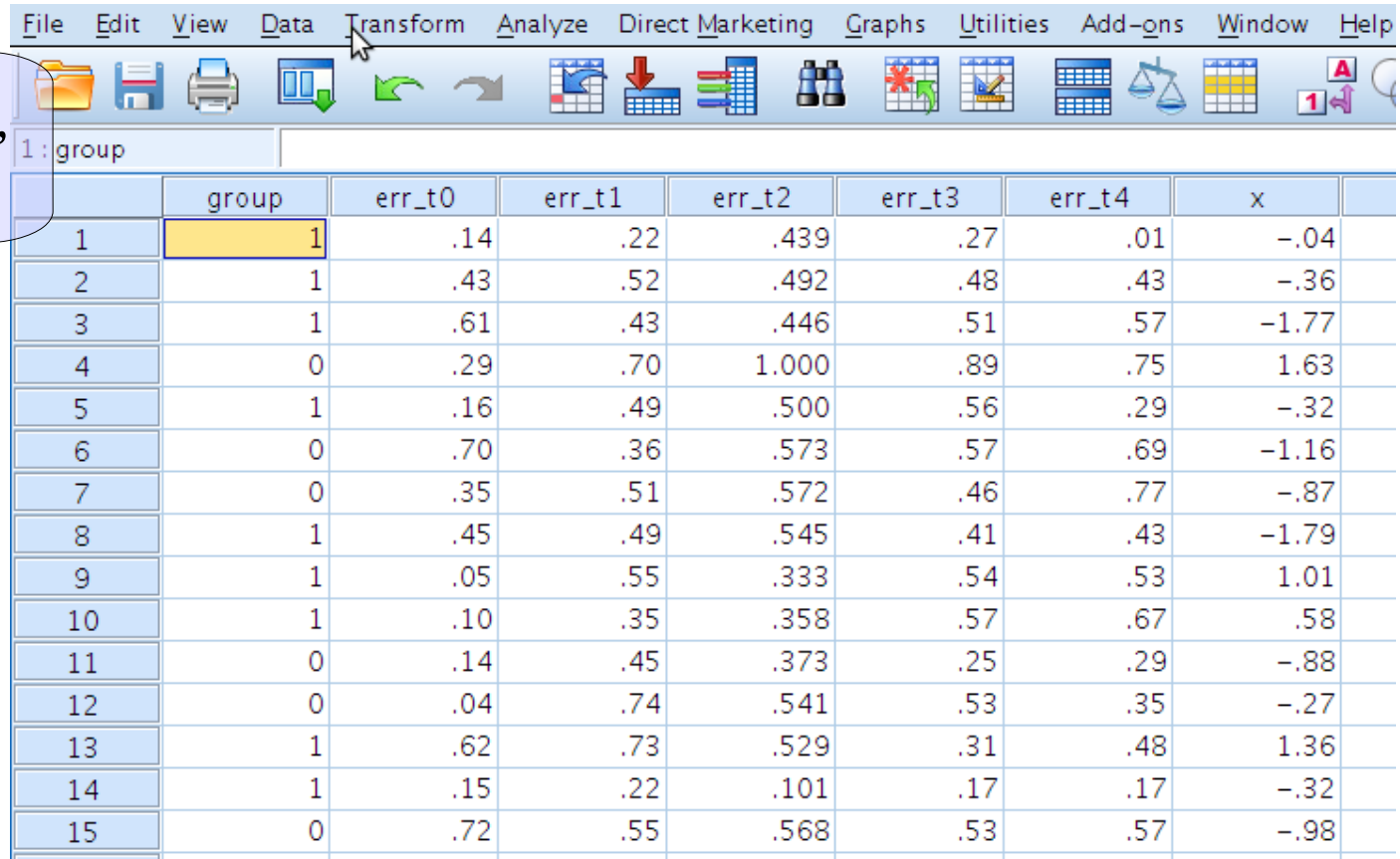
Condition

Participants	Condition				
	1	2	3	4	5
1	Y11	Y21	Y31	Y41	Y51
2	Y12	Y22	Y32	Y42	Y52
3	Y13	Y23	Y33	Y43	Y53
....					
N	Y1n	Y2n	Y3n	Y4n	Y5n

Standard file format

- As for many applications of the repeated-measure design, each level of the WS-factor is represented by a column in the file

One participant,
one row



	group	err_t0	err_t1	err_t2	err_t3	err_t4	x
1	1	.14	.22	.439	.27	.01	-.04
2	1	.43	.52	.492	.48	.43	-.36
3	1	.61	.43	.446	.51	.57	-1.77
4	0	.29	.70	1.000	.89	.75	1.63
5	1	.16	.49	.500	.56	.29	-.32
6	0	.70	.36	.573	.57	.69	-1.16
7	0	.35	.51	.572	.46	.77	-.87
8	1	.45	.49	.545	.41	.43	-1.79
9	1	.05	.55	.333	.54	.53	1.01
10	1	.10	.35	.358	.57	.67	.58
11	0	.14	.45	.373	.25	.29	-.88
12	0	.04	.74	.541	.53	.35	-.27
13	1	.62	.73	.529	.31	.48	1.36
14	1	.15	.22	.101	.17	.17	-.32
15	0	.72	.55	.568	.53	.57	-.98

Long file format

- For the mixed model we need to tabulate the data as if they came from a between-subject design

One measure,
one row

id	group	x	Condition	Y	value
1	1	-.037	1	.140	
1	1	-.037	2	.220	
1	1	-.037	3	.439	
1	1	-.037	4	.271	
1	1	-.037	5	.009	
2	1	-.360	1	.431	
2	1	-.360	2	.518	
2	1	-.360	3	.492	
2	1	-.360	4	.483	
2	1	-.360	5	.433	
3	1	-1.767	1	.612	
3	1	-1.767	2	.431	
3	1	-1.767	3	.446	
3	1	-1.767	4	.509	
3	1	-1.767	5	.573	
4	0	1.634	1	.291	
4	0	1.634	2	.702	

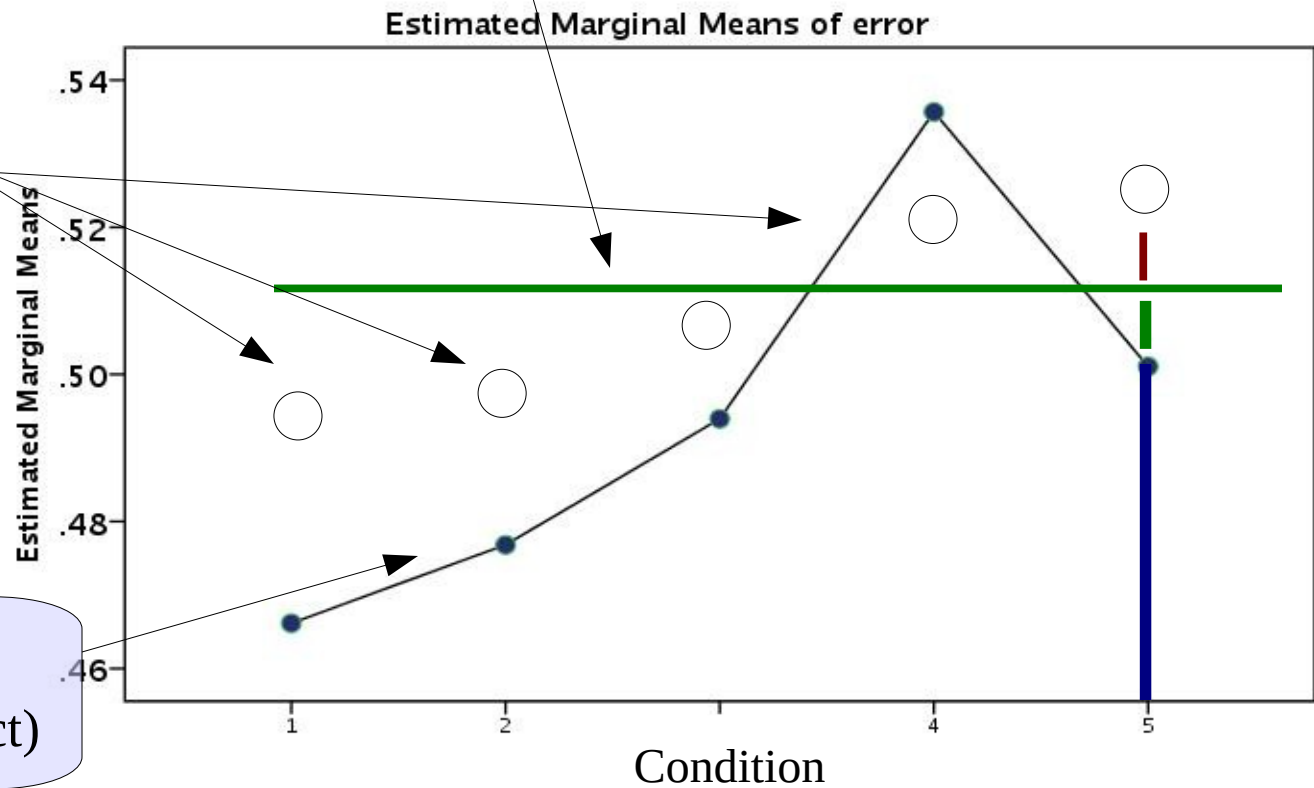
Participant scores

Plot for 1
participant

Participant
average trait

Participant
scores

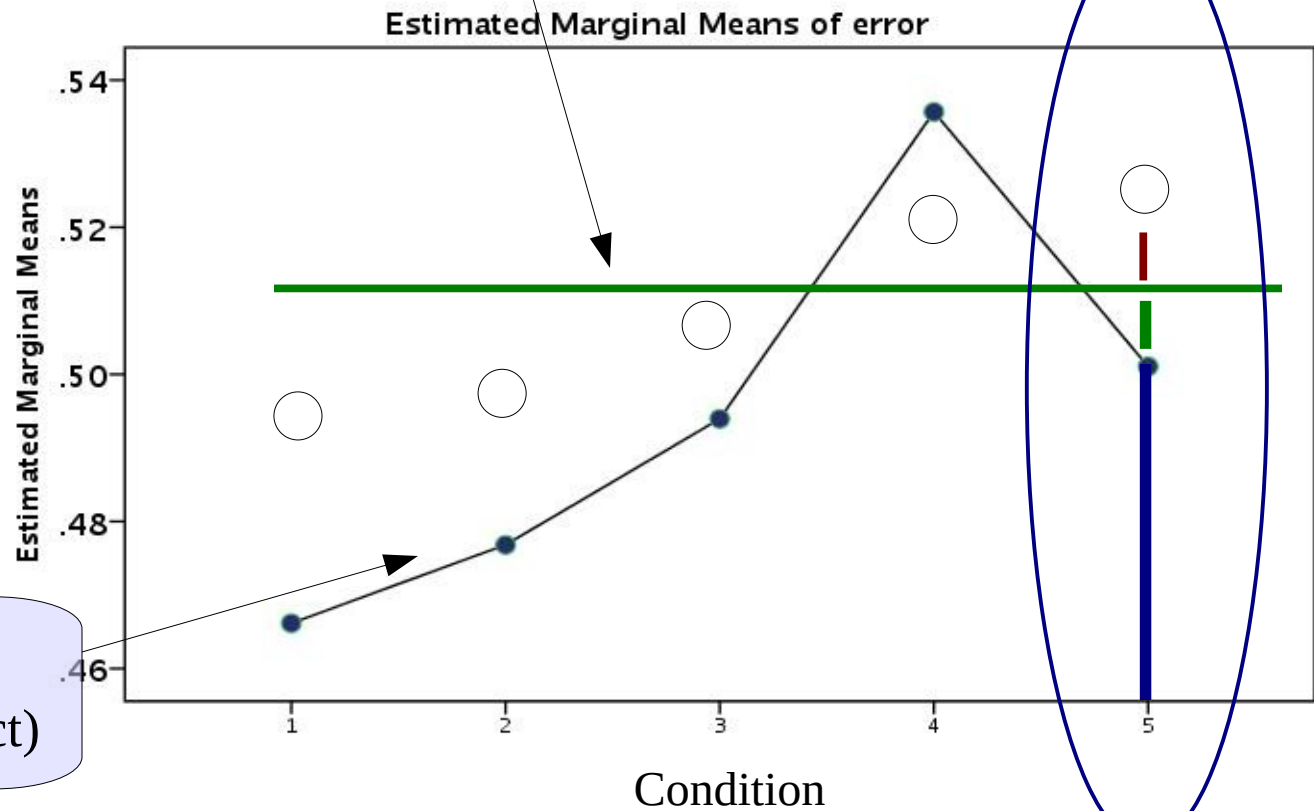
Averages of the
sample (fixed effect)



Where does the score come from?

Plot for 1 participant

Participant average trait



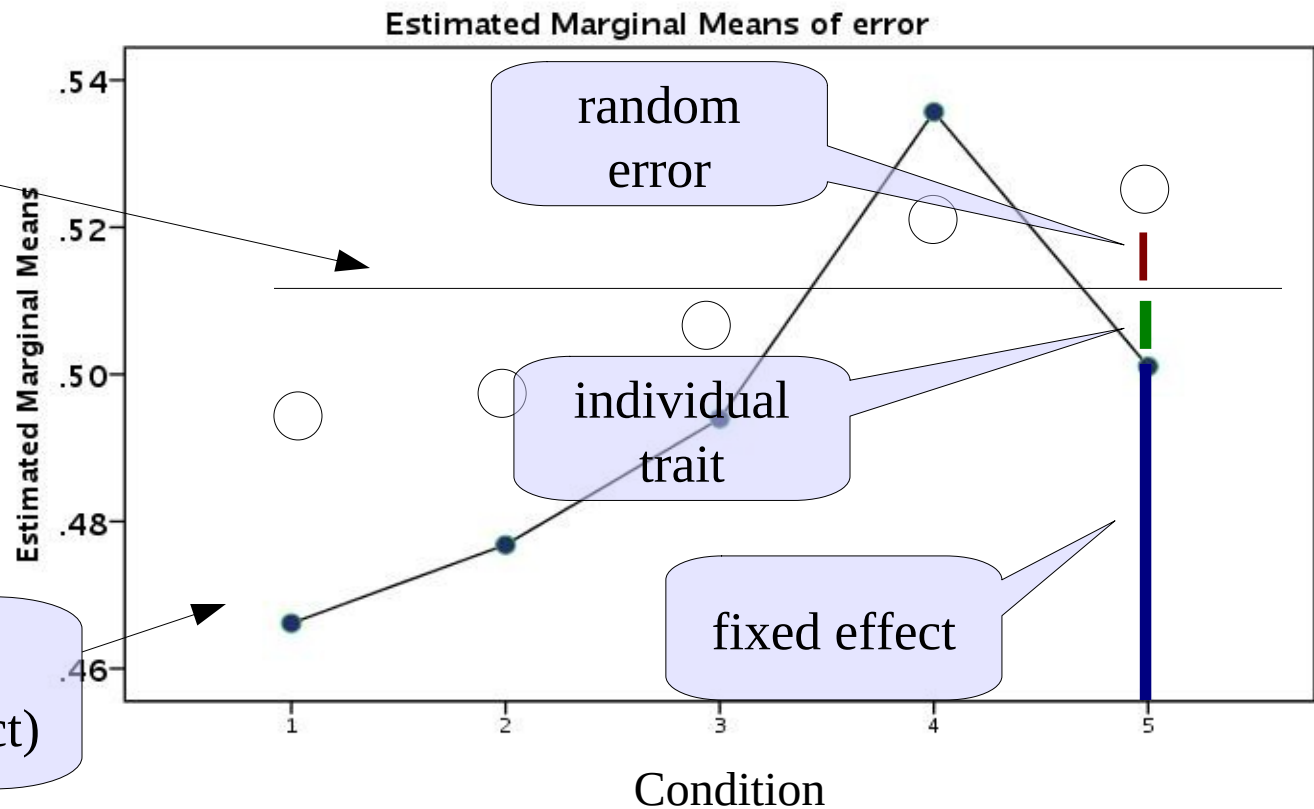
Averages of the sample (fixed effect)

Participant component

Plot for 1
participant

Participant
individual trait

Averages of the
sample (fixed effect)



Solution

Thus, we should consider an extra residual term which represents participants individual characteristic. This term is the same within each participant

$$Y_{11} = a + b_1 \cdot T_1 + u_1 + e_{11}$$

$$Y_{21} = a + b_2 \cdot T_2 + u_1 + e_{21}$$

$$Y_{31} = a + b_3 \cdot T_3 + u_1 + e_{31}$$

Effects of
condition

$$Y_{1j} = a + b_1 \cdot T_1 + u_j + e_{1j}$$

$$Y_{2j} = a + b_2 \cdot T_2 + u_j + e_{2j}$$

$$Y_{3j} = a + b_3 \cdot T_3 + u_j + e_{3j}$$

one participant
one trait

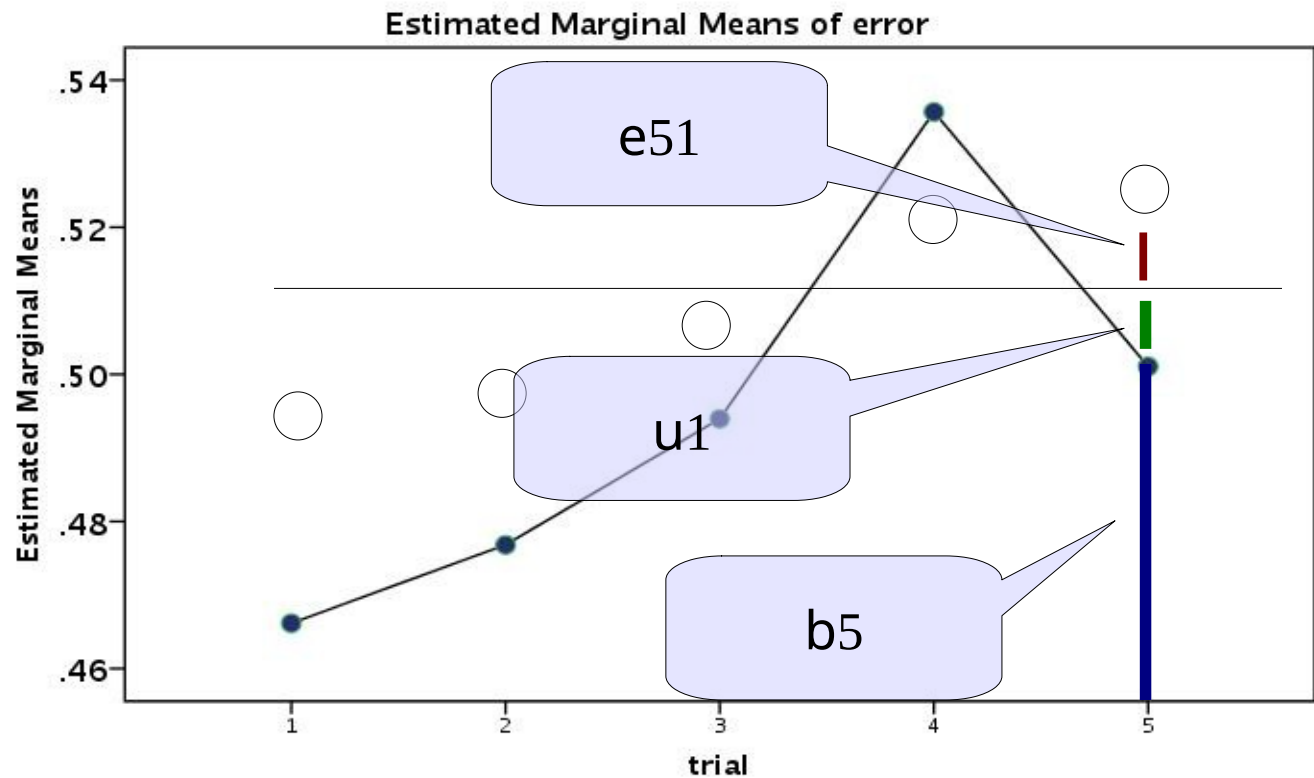
Each score,
one residual

Each score,
one error

One participant
one trait

Participant component

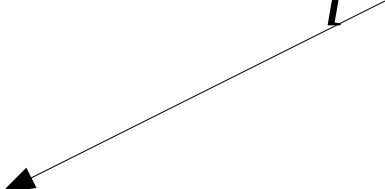
$$Y_{51} = a + b \cdot T_5 + u_1 + e_{51}$$



Building the model

We translate this in the standard mixed model

$$Y_{ij} = a + b' \cdot T_i + u_j + e_{ij}$$


$$y_{ij} = \bar{a} + a_j + \bar{b} \cdot x_{ij} + e_{ij}$$

- Fixed effects? Intercept and condition effect
- Random effects? Intercepts
- Clusters? participants

GAMLj: General mixed models

Categorical independent variable

Clustering variable(s)

Linear Mixed Model

group
x

Dependent Variable
y

Factors
condition

Covariates

Cluster variables
id

Estimation
☒ REML
☐ Do not run

Confidence Intervals
☒ Fixed parameters
☐ Random variances
Confidence level 95

Model comparison
☐ Activate
☐ More Fit Indices

GAMLj: random coefficients

Random intercepts

All possible random coefficients

Random Effects

Components

- Intercept | id
- condition | id

Random Coefficients

- Intercept | id

List components

☒ Model terms

Effects correlation

☒ Correlated

Tests

☐ LRT for Random Effects

GAMLj: fixed coefficients

Fixed effect

All possible fixed coefficients

Fixed Effects

Components

condition

Model Terms

condition

☒ Fixed Intercept

GAMLj: Results: model

R-squared

Model Results

Model Fit

Type	R ²	df	LRT X ²	p
Conditional	0.217	5	78.931	< .001
Marginal	0.015	4	18.773	< .001

[4]

R-squared Conditional: How much variance can the fixed and random effects together explain of the overall variance

R-squared Marginal: How much variance can the fixed effects alone explain of the overall variance

GAMLj: Results: random

Variance of intercepts

Random Components

Groups	Name	Variance	SD	ICC
id	(Intercept)	0.00780	0.0883	0.205
	Residual	0.03020	0.1738	

Note. Number of Obs: 1000 , Number of groups: id 200

As long as the variance is non-zero, we are fine

GAMLj: Results: fixed

F-tests

Fixed Effects Omnibus Tests

	F	df	df (res)	p
condition	4.72	4	796	< .001

Coefficients

Parameter Estimates (Fixed coefficients)

Names	Effect	Estimate	SE	95% Confidence Intervals		df	t	p
				Lower	Upper			
(Intercept)	(Intercept)	0.4947	0.00832	0.47841	0.5111	199	59.462	< .001
condition1	2 - 1	0.0107	0.01738	-0.02344	0.0448	796	0.613	0.540
condition2	3 - 1	0.0278	0.01738	-0.00632	0.0619	796	1.598	0.110
condition3	4 - 1	0.0695	0.01738	0.03541	0.1036	796	4.000	< .001
condition4	5 - 1	0.0349	0.01738	8.03e-4	0.0690	796	2.009	0.045

Contrasts used to cast
the categorical IV

GAMLj: plot

Fixed effect to plot

Options

Plots

Horizontal axis

→ condition

Separate lines

→

Separate plots

→

Display

☐ None

☒ Confidence intervals

☐ Standard Error

Plot

☐ Observed scores

☐ Y-axis observed range

Use

☐ Original scale

☐ Varying line types

Random Effects

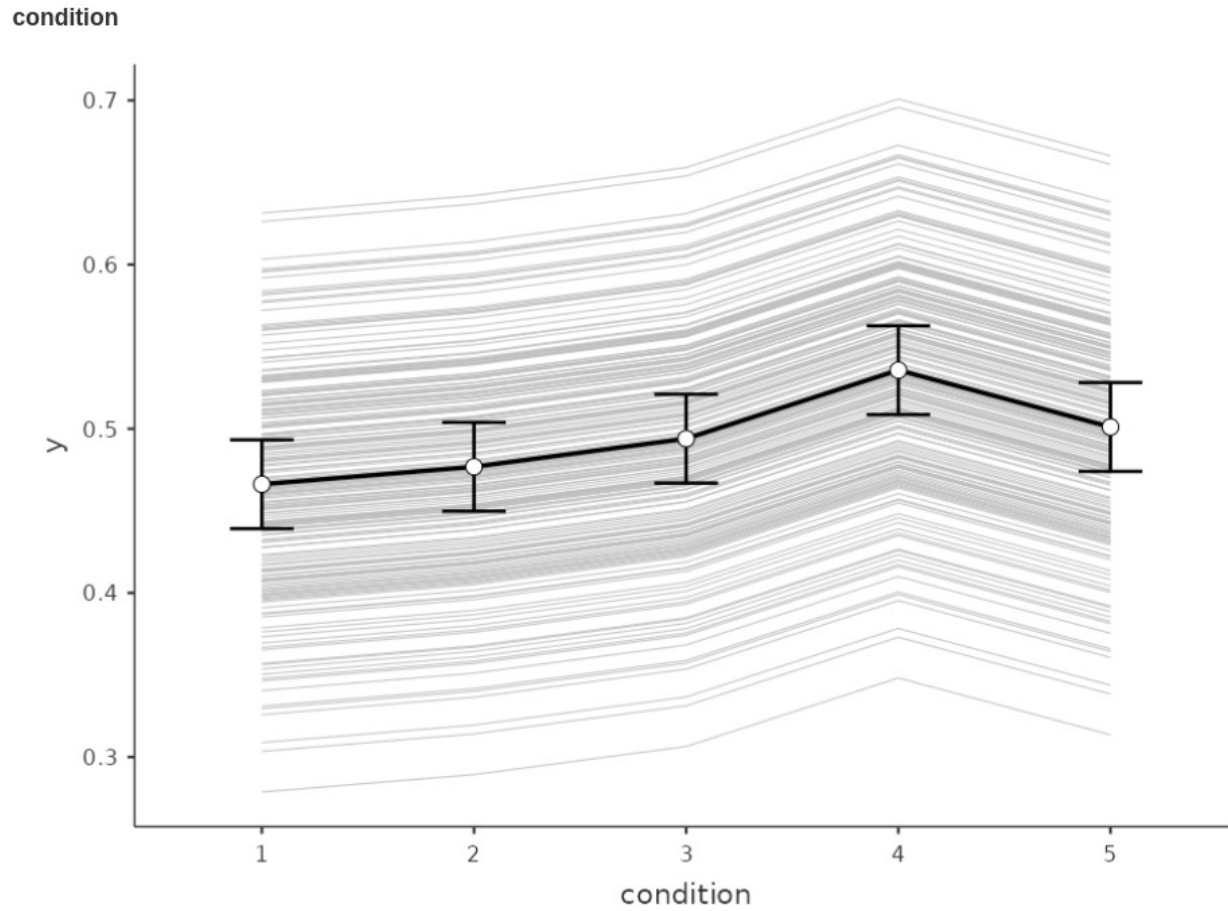
☒ Random effects

☒ Average method

☐ Full predicted method

GAMLj: plot

Results Plots



GAMLj: post-hoc

- As in GLM (Anova), sometimes we want to compare conditions using post-hoc tests. GAMLj allows for Bonferroni and Tukey (more liberal) p-value adjustment

The screenshot shows the 'Post Hoc Tests' dialog box in GAMLj. At the top, there is a dropdown menu labeled 'Post Hoc Tests'. Below this, there is a large empty box on the left and a box on the right containing the text 'condition'. An arrow points from the left box to the right box. At the bottom, there are two sections: 'Correction' and 'Info'. The 'Correction' section has a list of checkboxes: 'No correction (LSD)', 'Bonferroni' (which is checked), 'Tukey', 'Holm', 'Scheffe', and 'Sidak'. The 'Info' section has a checkbox for 'Confidence Intervals'.

Post Hoc Tests

condition

Correction

- ☐ No correction (LSD)
- ☒ Bonferroni
- ☐ Tukey
- ☐ Holm
- ☐ Scheffe
- ☐ Sidak

Info

- ☐ Confidence Intervals

GAMLj: post-hoc

- The interpretation follows as for any standard ANOVA

Post Hoc Tests

Post Hoc Comparisons - trial

Comparison		Difference	SE	t	df	Pbonferroni	Pholm
trial	trial						
1	- 2	-0.01066	0.0174	-0.613	796	1.000	1.000
	- 3	-0.02778	0.0174	-1.598	796	1.000	0.552
	- 4	-0.06951	0.0174	-4.000	796	< .001	< .001
	- 5	-0.03491	0.0174	-2.009	796	0.449	0.314
2	- 3	-0.01712	0.0174	-0.985	796	1.000	0.975
	- 4	-0.05885	0.0174	-3.386	796	0.007	0.007
	- 5	-0.02425	0.0174	-1.395	796	1.000	0.653
3	- 4	-0.04173	0.0174	-2.401	796	0.166	0.133
	- 5	-0.00713	0.0174	-0.410	796	1.000	1.000
4	- 5	0.03460	0.0174	1.991	796	0.468	0.314

Between and Repeated Measures Anova

linear mixed model

Standard design

- There are two groups - a Control group and a Treatment group, measured at 4 times. These times are labeled as 1 (pretest), 2 (one month posttest), 3 (3 months follow-up), and 4 (6 months follow-up).
- The dependent variable is a depression score (e.g. Beck Depression Inventory) and the treatment is drug versus no drug. If the drug worked about as well for all subjects the slopes would be comparable and negative across time. For the control group we would expect some subjects to get better on their own and some to stay depressed, which would lead to differences in slope for that group (*)

Standard design

- There are two groups - a Control group and a Treatment group, measured at 4 times. These times are labeled as 1 (pretest), 2 (one month posttest), 3 (3 months follow-up), and 4 (6 months follow-up).

Contingency Tables

Contingency Tables

time	group		Total
	1	2	
0	12	12	24
1	12	12	24
3	12	12	24
6	12	12	24
Total	48	48	96

96 observations
24 subjects

Standard design: data

- Data are in the long format

One subject 4 rows

	subj	time	group	dv	
1	1	0	1	296	
2	1	1	1	175	
3	1	3	1	187	
4	1	6	1	192	
5	2	0	1	376	
6	2	1	1	329	
7	2	3	1	236	
8	2	6	1	76	
9	3	0	1	309	
10	3	1	1	238	
11	3	3	1	150	
12	3	6	1	123	
13	4	0	1	222	
14	4	1	1	60	
15	4	3	1	82	
16	4	6	1	85	
17	5	0	1	150	
18	5	1	1	271	
19	5	3	1	250	

Mixed model

We can translate this in a standard mixed model

- Fixed effects? Intercept and group,time, and interaction effect
- Random effects? Intercepts
- Clusters? subjects

Variables

- Definition of the analysis

Clustering variable

Mixed Model

Dependent Variable
→ dv

Factors
→ time
group

Covariates
→

Cluster variables
→ subj

Estimation
☒ REML

Confidence Intervals
☒ Confidence intervals Interval 95 %

Model

- Definition of the analysis

Fixed effects

Fixed Effects

Components

time
group

Model Terms

time
group
time * group

☒ Fixed Intercept

Random effects

Random Effects

Components

time | subj
group | subj
time : group | subj

Random Coefficients

Intercept | subj

☒ Correlated Effects

Results

- Interpretation of results

Model

Model Results

Model Fit

Type	R ²	df	LRT X ²	p
Conditional	0.768	8	107.251	< .001
Marginal	0.554	7	101.043	< .001

[4]

Random effects

Random Components

Groups	Name	Variance	SD	ICC
subj	(Intercept)	2539	50.4	0.479
Residual		2761	52.5	

Note. Number of Obs: 96 , Number of groups: subj 24

Results

- Interpretation of results

Fixed F-tests

Fixed Effect ANOVA

	F	Num df	Den df	p
time	45.14	3	66.0	< .001
group	13.71	1	22.0	0.001
time:group	9.01	3	66.0	< .001

Note. Satterthwaite method for degrees of freedom

- For the moment we ignore the coefficients of the parameter estimates

Results: plot

- Interpretation of results

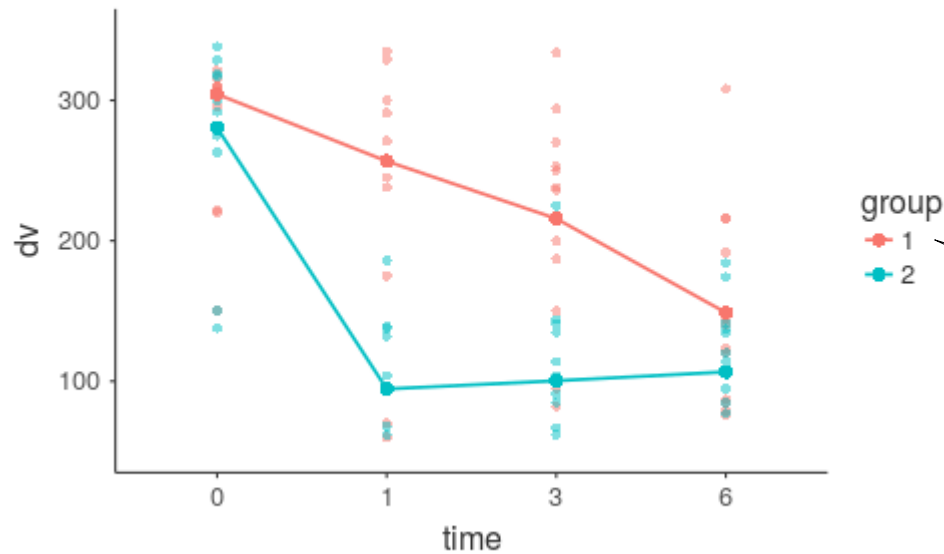
Fixed Effects Plots

Horizontal axis
→ time

Separate lines
→ group

Separate plots
→

Fixed Effects Plots



Red is the control group

Probing the results

- We can probe the interaction (and the pattern of means) in different ways (all available in GAMLj):
- Simple effects: Test if the effects of time is there (and how strong it is) for different groups
- Post-hoc test

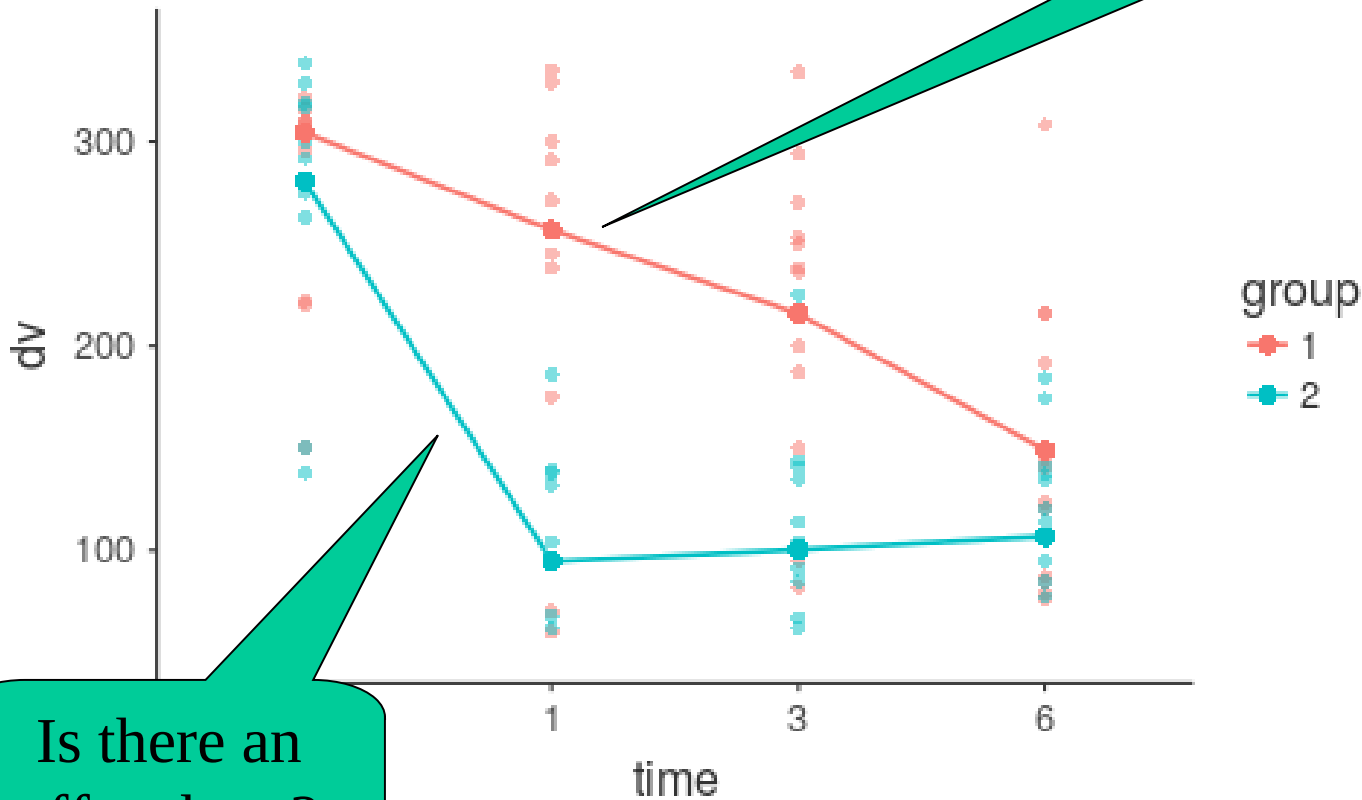
Simple effect analysis

Simple effects

- Interpretation of results

Fixed Effects Plots

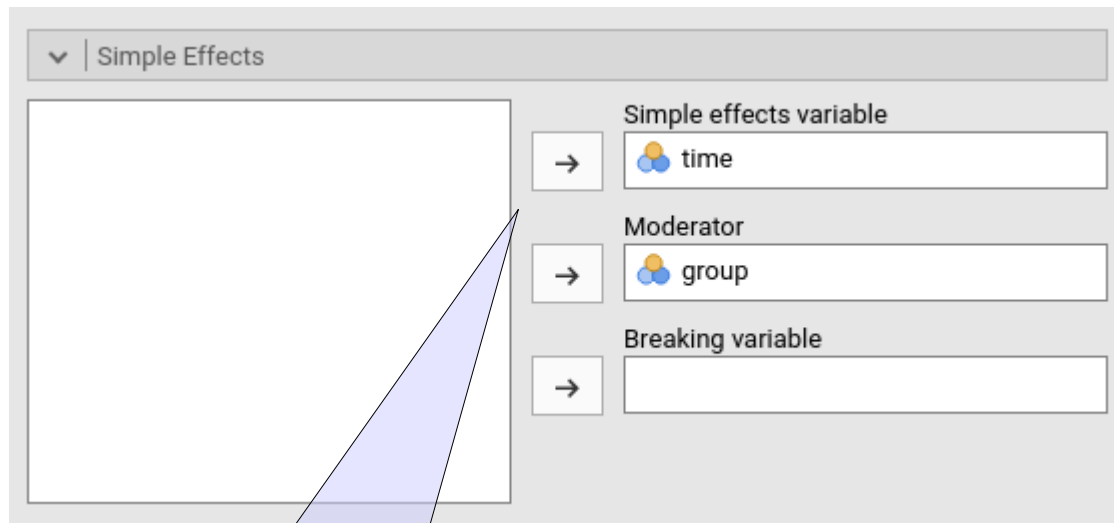
Is there an effect here?



Is there an effect here?

Simple effects

- We should declare which is the variable we want the effect for and which is the moderator



The screenshot shows a dialog box titled "Simple Effects". On the left is a large empty rectangular area. On the right, there are three input fields, each preceded by a right-pointing arrow in a small box. The first field is labeled "Simple effects variable" and contains the text "time". The second field is labeled "Moderator" and contains the text "group". The third field is labeled "Breaking variable" and is currently empty.

Effects of time for
different groups

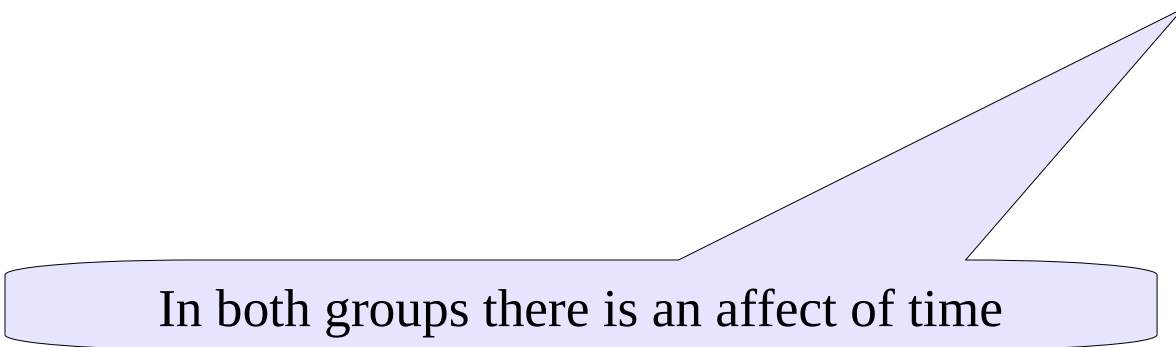
Simple effects

- We can say that the treatment works for both groups, although in a different way (recall the interaction)

Simple Effects ANOVA

Simple effects of time

Effect	Moderator Levels	df Num	df Den	F	p
time	group at 1	3.00	66.0	18.9	< .001
time	group at 2	3.00	66.0	35.3	< .001



In both groups there is an affect of time

Cross-classified random factors:
More than one clustering variable

Subject by stimuli design

- Several modern research designs have more than one clustering variable:

- Repeated measures within participants with several stimuli

An example of stimuli crossed with conditions can be: A researcher investigates the effect of colors on reading words. A list of words is randomly chosen from the dictionary and it is used as the set of stimuli in the experiment. The experimental factor is color of the word presented to the participant, repeated within participants: each word is presented to the participant once in blue and once in red on the computer screen. Thus, each participant reads all words twice.

- Assume we have $N=50$ subjects and $S=30$ stimuli. 2 conditions

Subject by stimuli design

Contingency Tables

Contingency Tables

subj	cond		Total
	A	B	
1	30	30	60
2	30	30	60
3	30	30	60
4	30	30	60
5	30	30	60
6	30	30	60
7	30	30	60
8	30	30	60
9	30	30	60
10	30	30	60
11	30	30	60
12	30	30	60
13	30	30	60
14	30	30	60
15	30	30	60
16	30	30	60
17	30	30	60

.....

Each subject has 60 measures, 30 stimuli per condition

Each subject may have different mean in the dependent variable: random intercept across subjects

Each subject may have different effect of condition (A-B): random slopes

Subject by stimuli design

Contingency Tables

Contingency Tables

stimulus	cond		Total
	A	B	
1	50	50	100
2	50	50	100
3	50	50	100
4	50	50	100
5	50	50	100
6	50	50	100
7	50	50	100
8	50	50	100
9	50	50	100
10	50	50	100
11	50	50	100
12	50	50	100

.....

Each stimulus has 100 measures,
50 subjects per condition

Each stimulus may have
different mean in the dependent
variable: random intercept across
stimuli

Each stimulus may have
different effect of condition (A-
B): random slopes across stimuli

Building the model

- Clusters? Subjects and Stimuli
- Fixed effects? Intercept and condition
- Random effects?
 - Intercepts and condition across subjects
 - Intercepts and condition across stimuli

Cross-classified mixed model

Building the model

- Let's do the analyses together with **jamovi**
- Data are “subjects_by_stimuli” example dataset